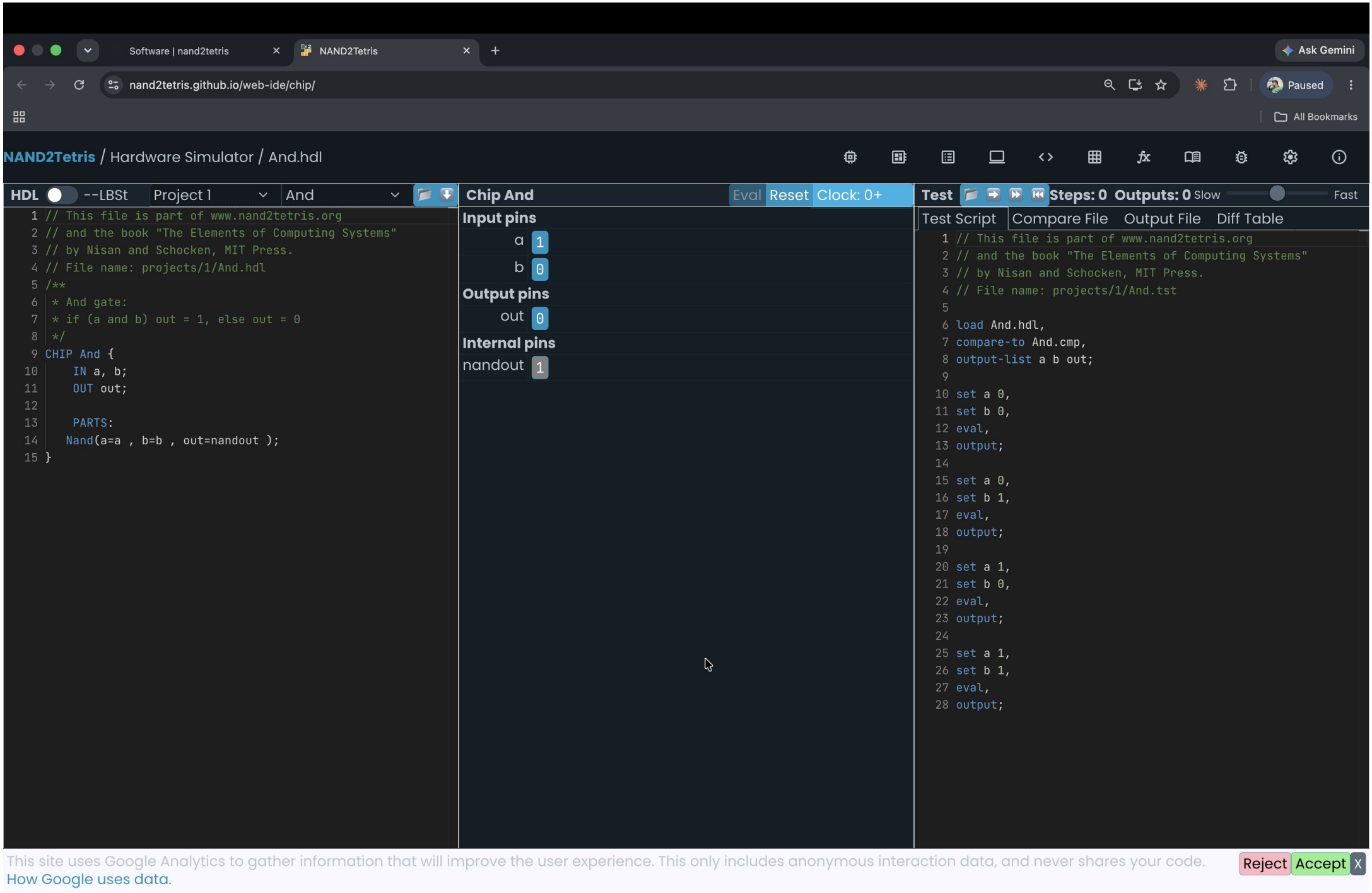
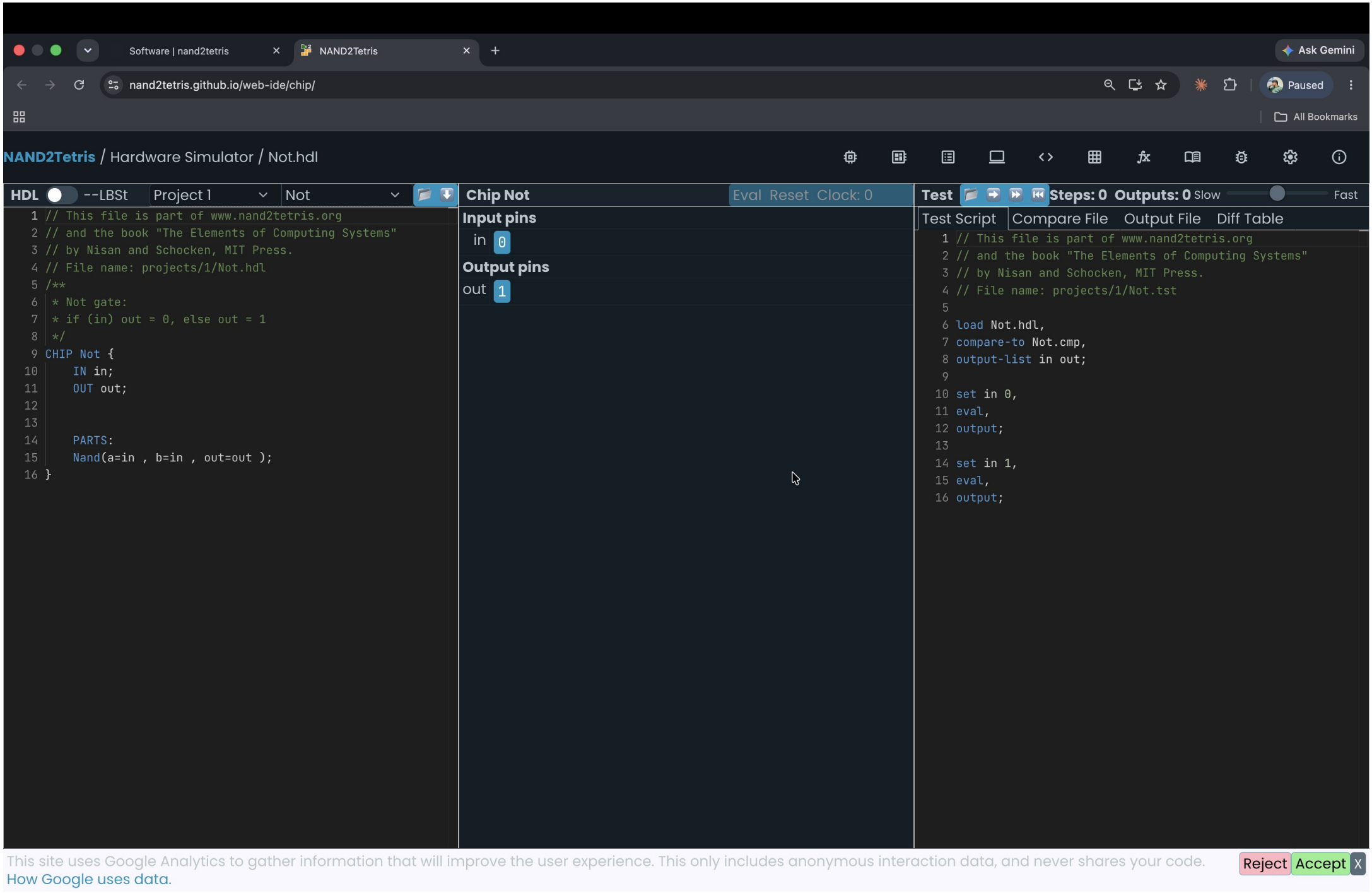
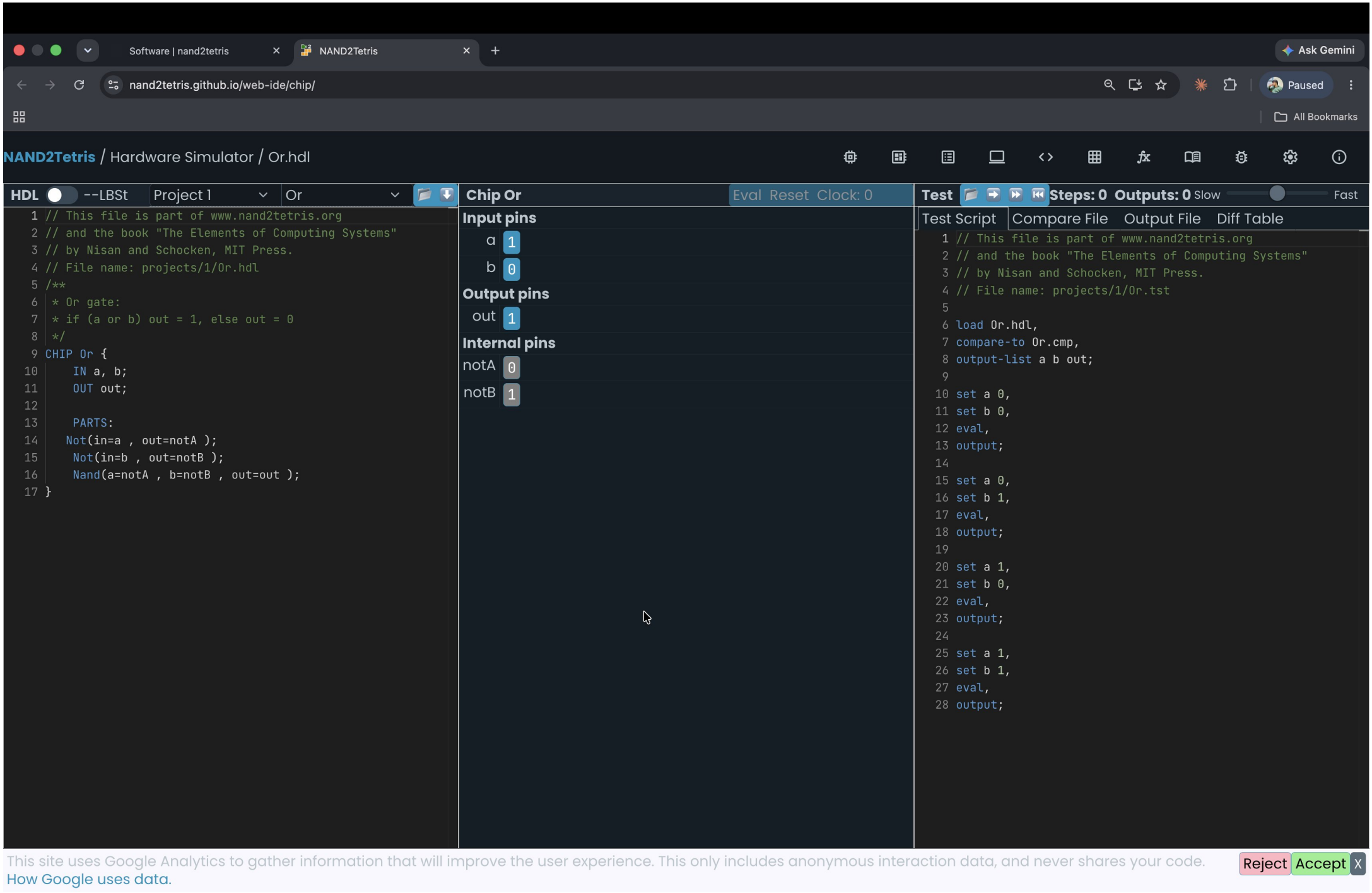
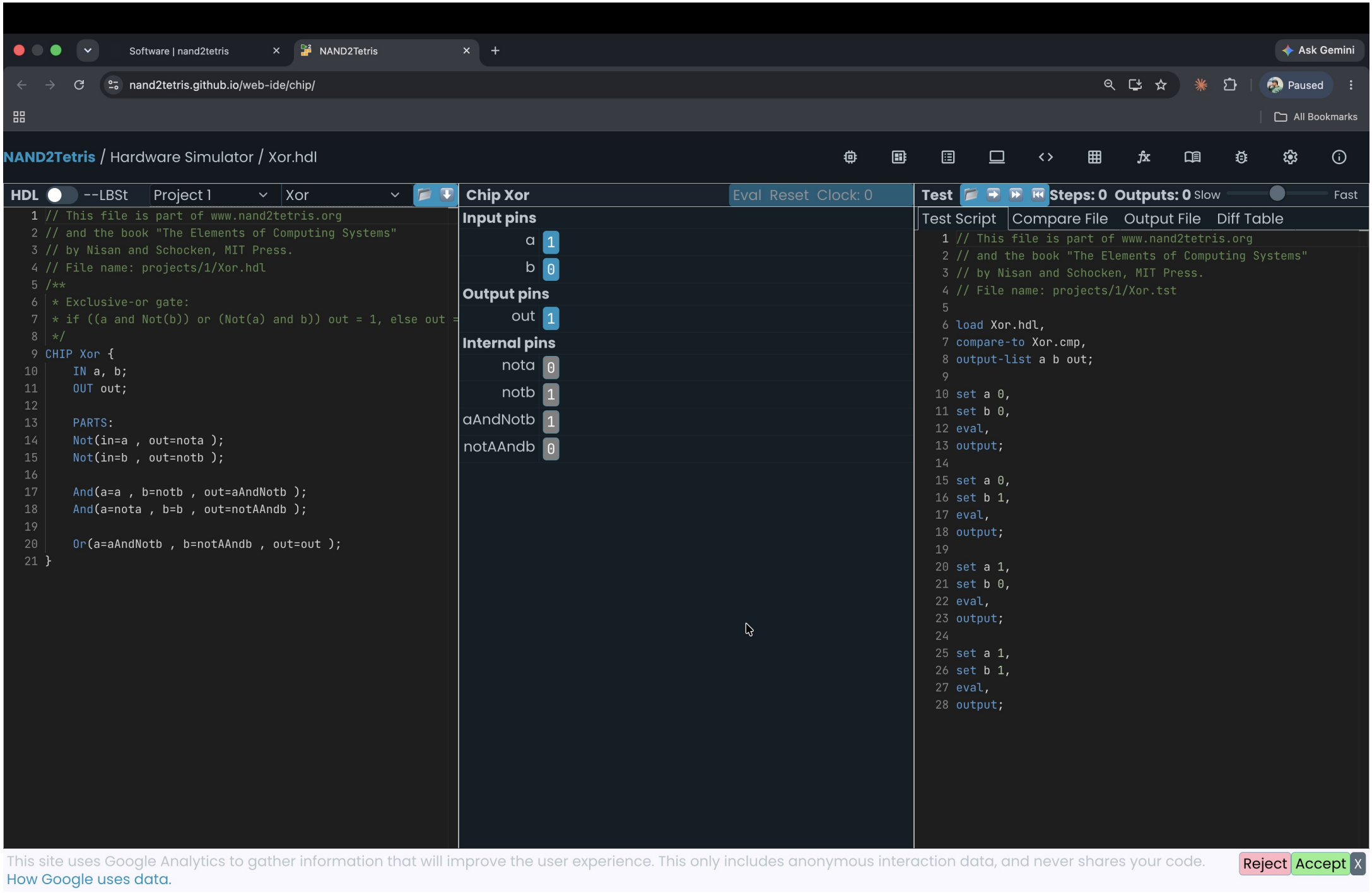
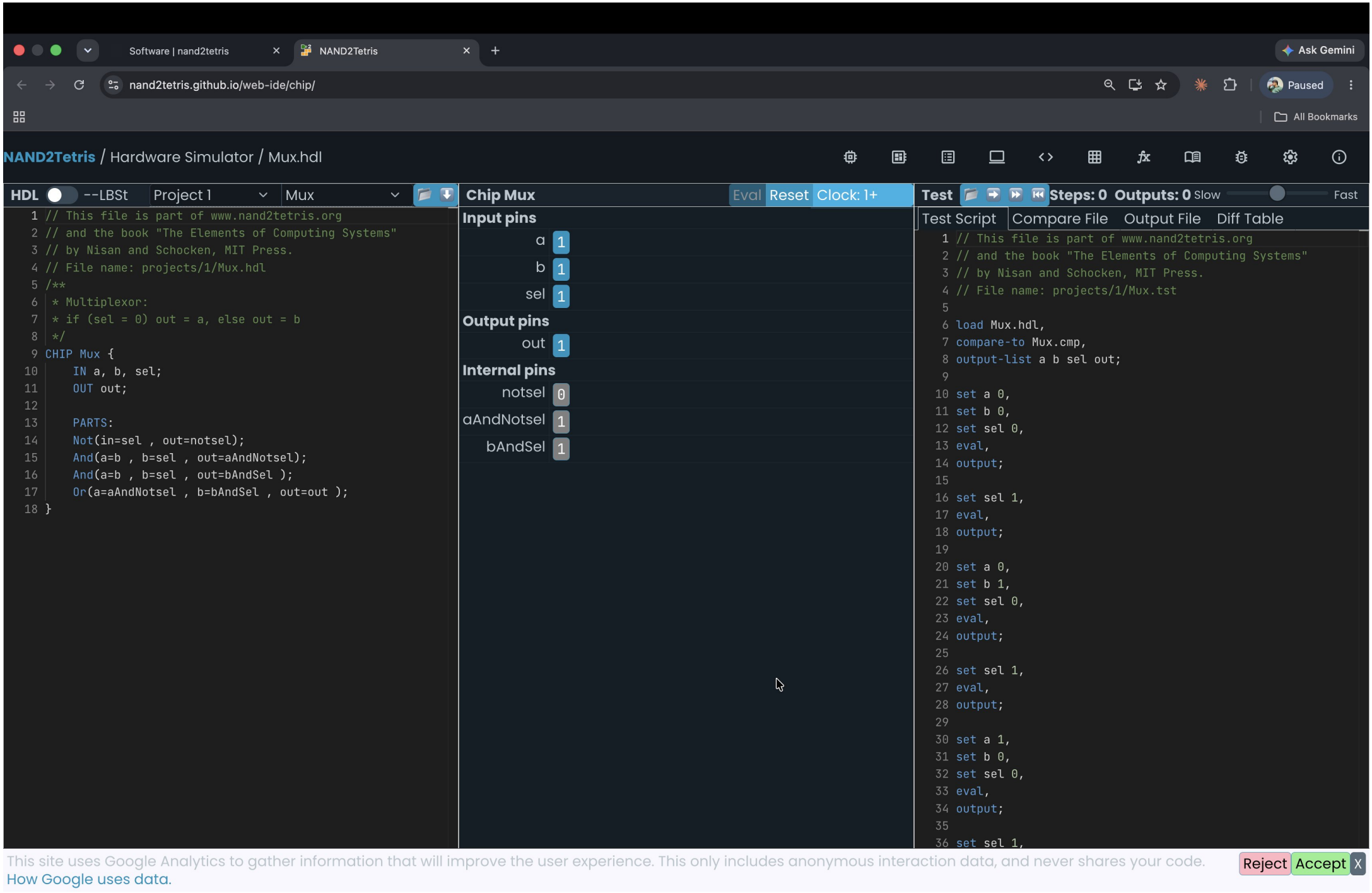

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NAND2Tetris / Hardware Simulator / And.hdl
HDL --LBSt Project 1 And
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/And.hdl
/**
* And gate:
* if (a and b) out = 1, else out = 0
*/
CHIP And {
IN a, b;
OUT out;
PARTS:
Nand(a=a , b=b , out=nandout );
}
Chip And Eval Reset Clock: 0+
Input pins
a 1
b 0
Output pins
out 0
Internal pins
nandout 1
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/And.tst
load And.hdl,
compare-to And.cmp,
output-list a b out;
set a 0,
set b 0,
eval,
output;
set a 0,
set b 1,
eval,
output;
set a 1,
set b 0,
eval,
output;
set a 1,
set b 1,
eval,
output;
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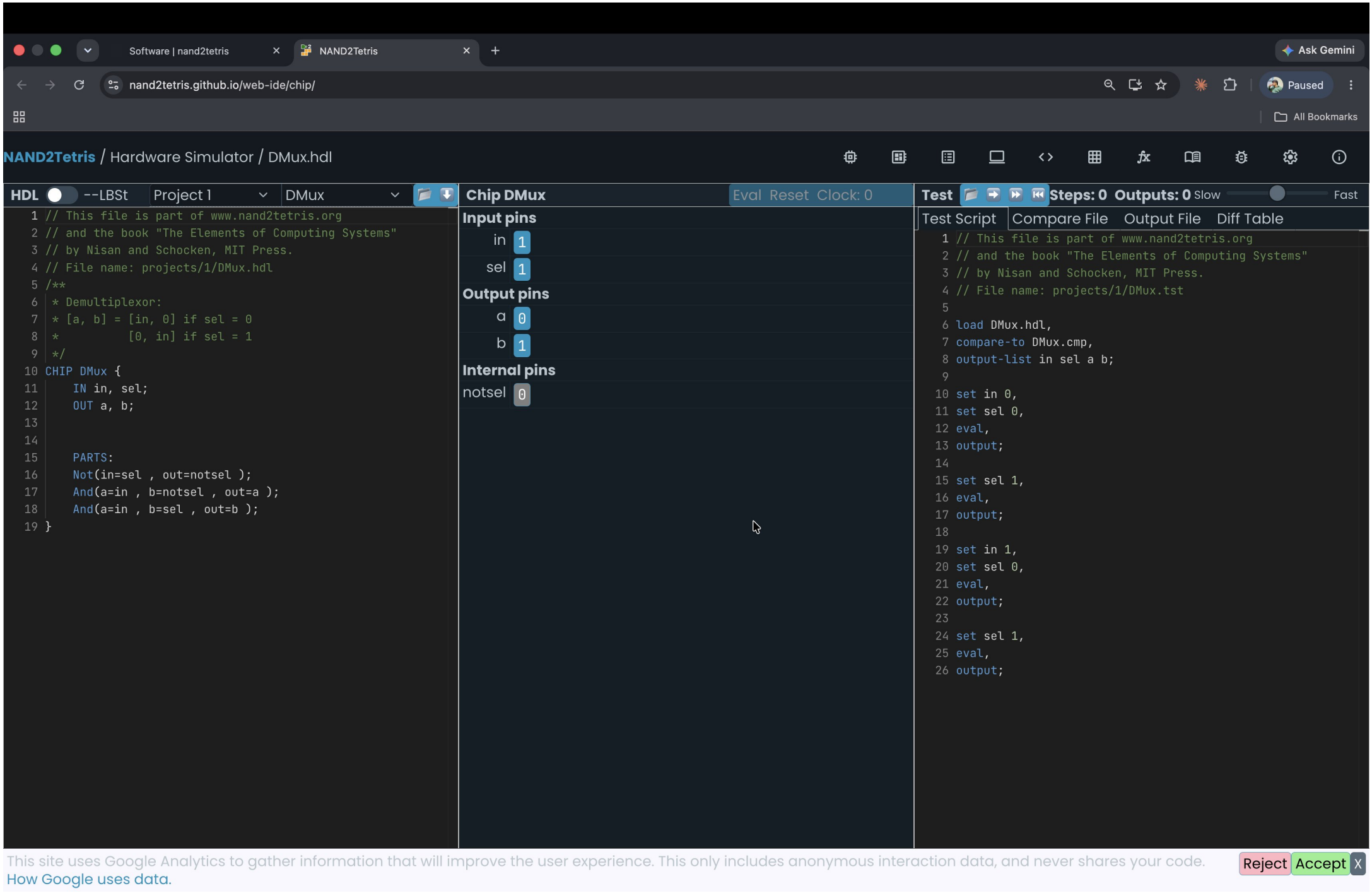

Software | nand2tetris
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NAND2Tetris / Hardware Simulator / Not.hdl
HDL --LBSt Project 1 Not
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Not.hdl
/**
* Not gate:
* if (in) out = 0, else out = 1
*/
CHIP Not {
IN in;
OUT out;
PARTS:
Nand(a=in , b=in , out=out );
}
Chip Not
Eval Reset Clock: 0
Input pins
in 0
Output pins
out 1
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Not.tst
load Not.hdl,
compare-to Not.cmp,
output-list in out;
set in 0,
eval,
output;
set in 1,
eval,
output;
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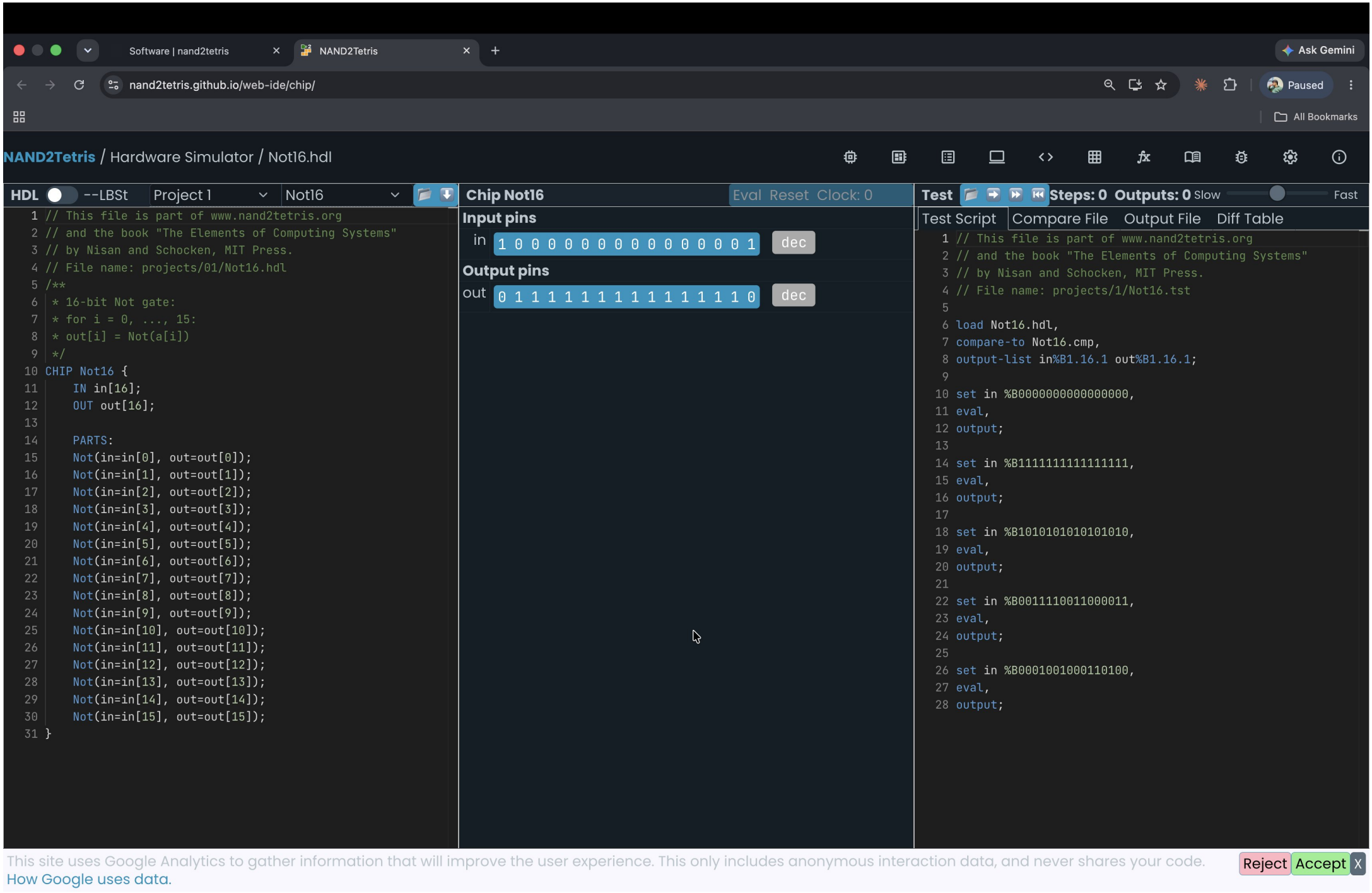


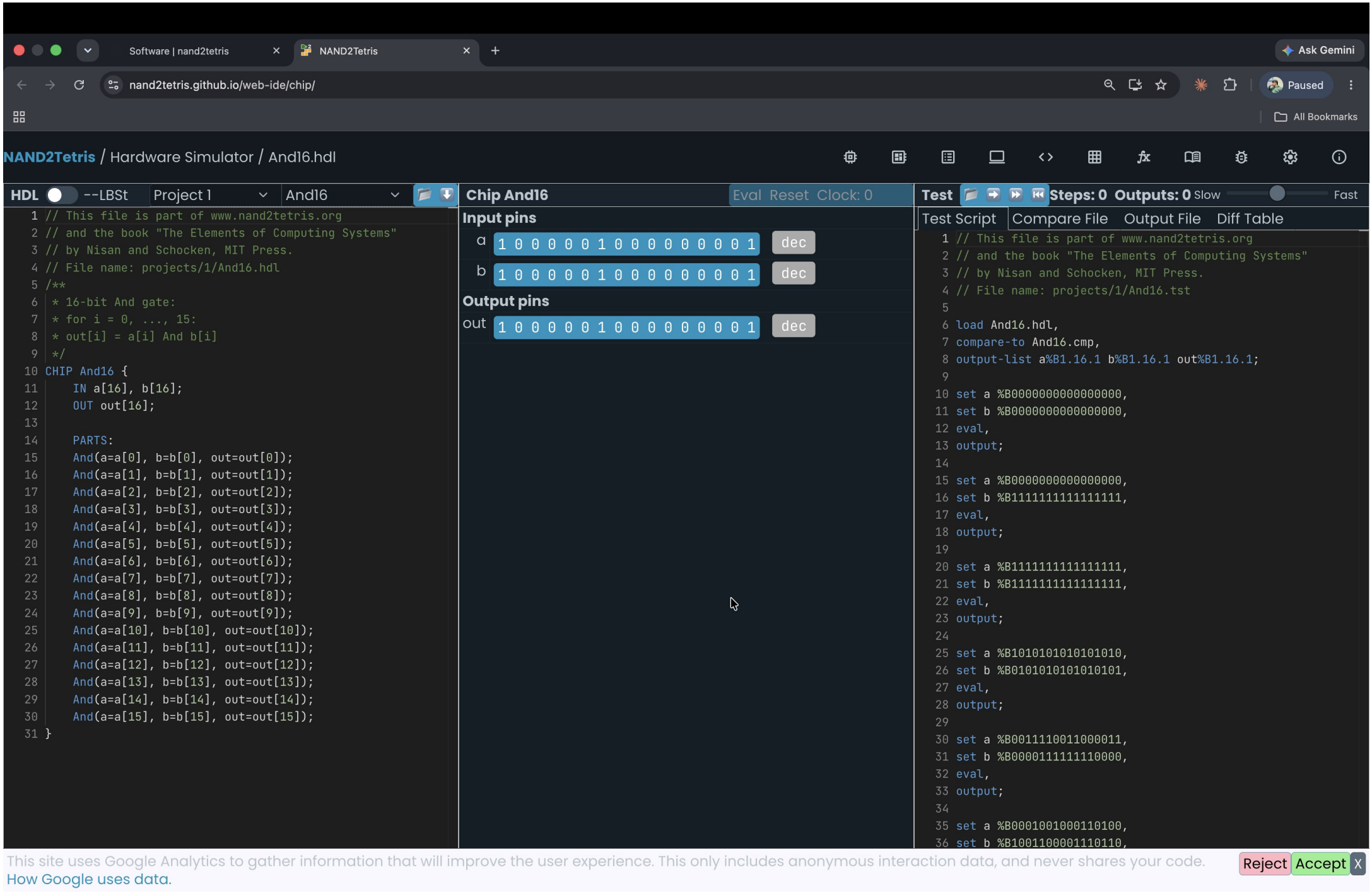

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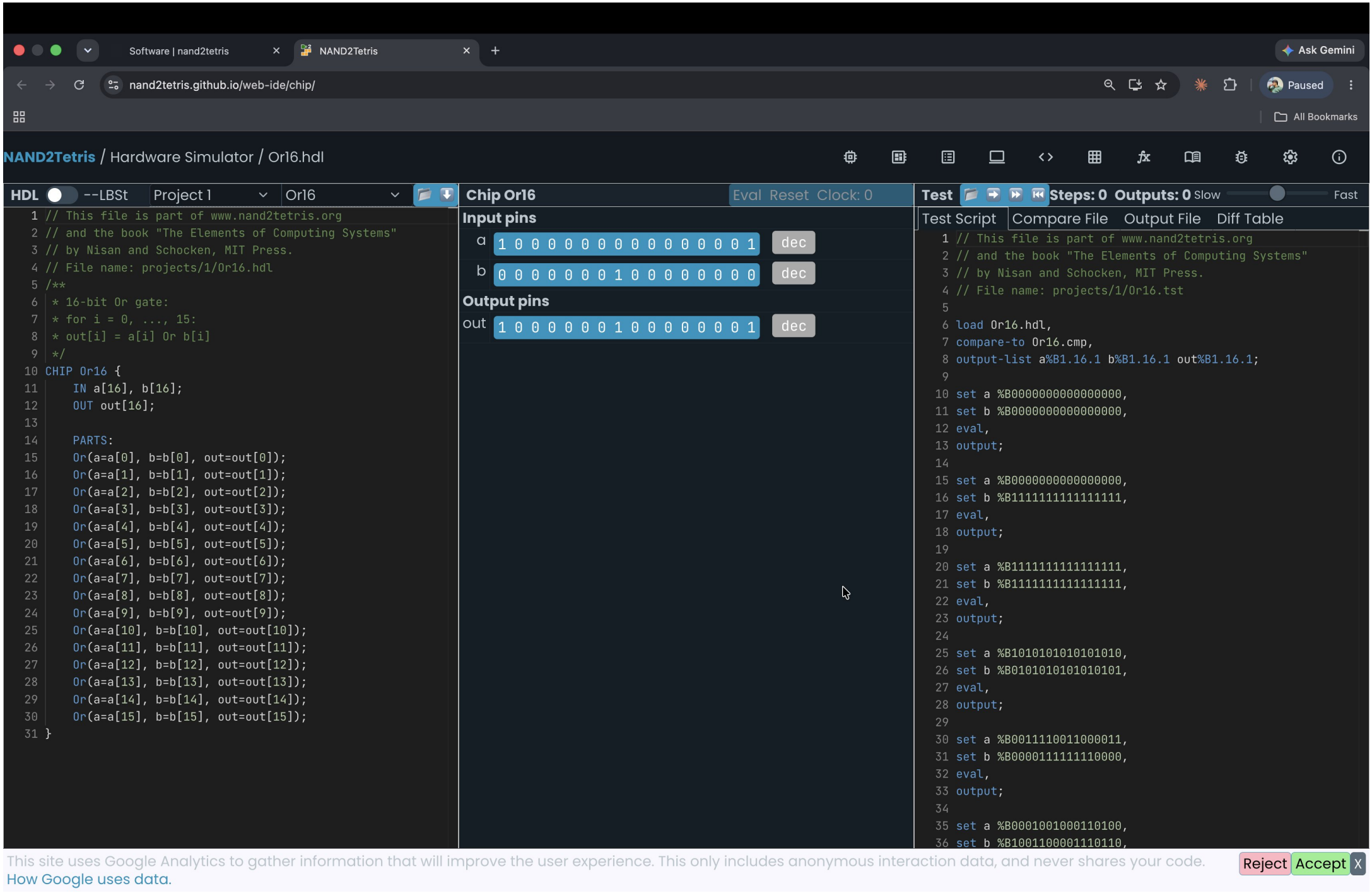

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NAND2Tetris / Hardware Simulator / Xor.hdl
HDL --LBSt Project 1 Xor
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Xor.hdl
/**
* Exclusive-or gate:
* if ((a and Not(b)) or (Not(a) and b)) out = 1, else out =
*/
CHIP Xor {
IN a, b;
OUT out;
PARTS:
Not(in=a , out=nota );
Not(in=b , out=notb );
And(a=a , b=notb , out=aAndNotb );
And(a=nota , b=b , out=notAAndb );
Or(a=aAndNotb , b=notAAndb , out=out );
}
Chip Xor
Eval Reset Clock: 0
Input pins
a 1
b 0
Output pins
out 1
Internal pins
nota 0
notb 1
aAndNotb 1
notAAndb 0
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Xor.tst
load Xor.hdl,
compare-to Xor.cmp,
output-list a b out;
set a 0,
set b 0,
eval,
output;
set a 0,
set b 1,
eval,
output;
set a 1,
set b 0,
eval,
output;
set a 1,
set b 1,
eval,
output;
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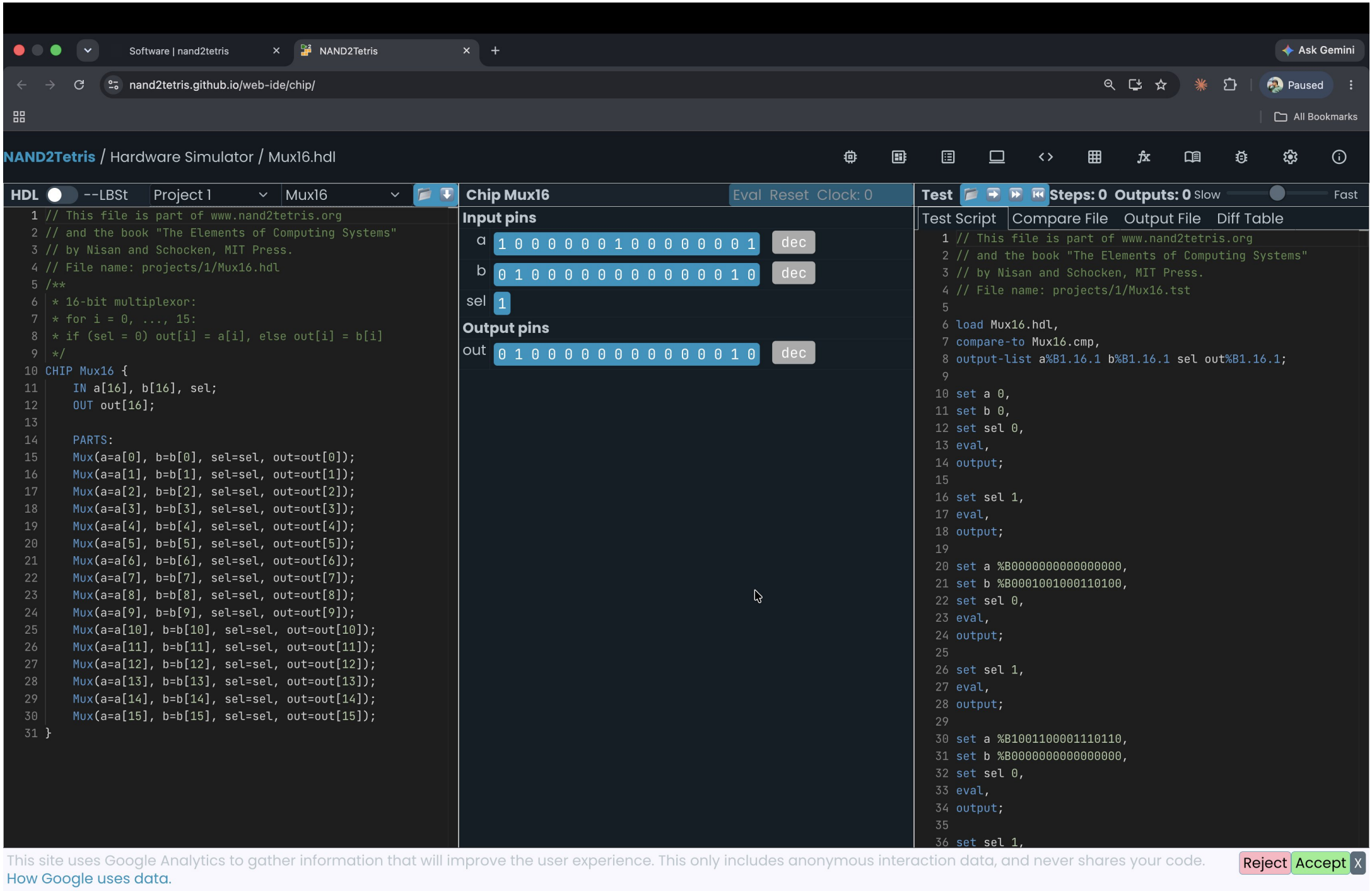

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NAND2Tetris / Hardware Simulator / Mux.hdl
HDL --LBSt Project 1 Mux
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Mux.hdl
/**
* Multiplexor:
* if (sel = 0) out = a, else out = b
*/
CHIP Mux {
IN a, b, sel;
OUT out;
PARTS:
Not(in=sel , out=notsel);
And(a=b , b=sel , out=aAndNotsel);
And(a=b , b=sel , out=bAndSel );
Or(a=aAndNotsel , b=bAndSel , out=out );
}
Chip Mux Eval Reset Clock: 1+
Input pins
a 1
b 1
sel 1
Output pins
out 1
Internal pins
notsel 0
aAndNotsel 1
bAndSel 1
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Mux.tst
load Mux.hdl,
compare-to Mux.cmp,
output-list a b sel out;
set a 0,
set b 0,
set sel 0,
eval,
output;
set sel 1,
eval,
output;
set a 0,
set b 1,
set sel 0,
eval,
output;
set sel 1,
eval,
output;
set a 1,
set b 0,
set sel 0,
eval,
output;
set sel 1,
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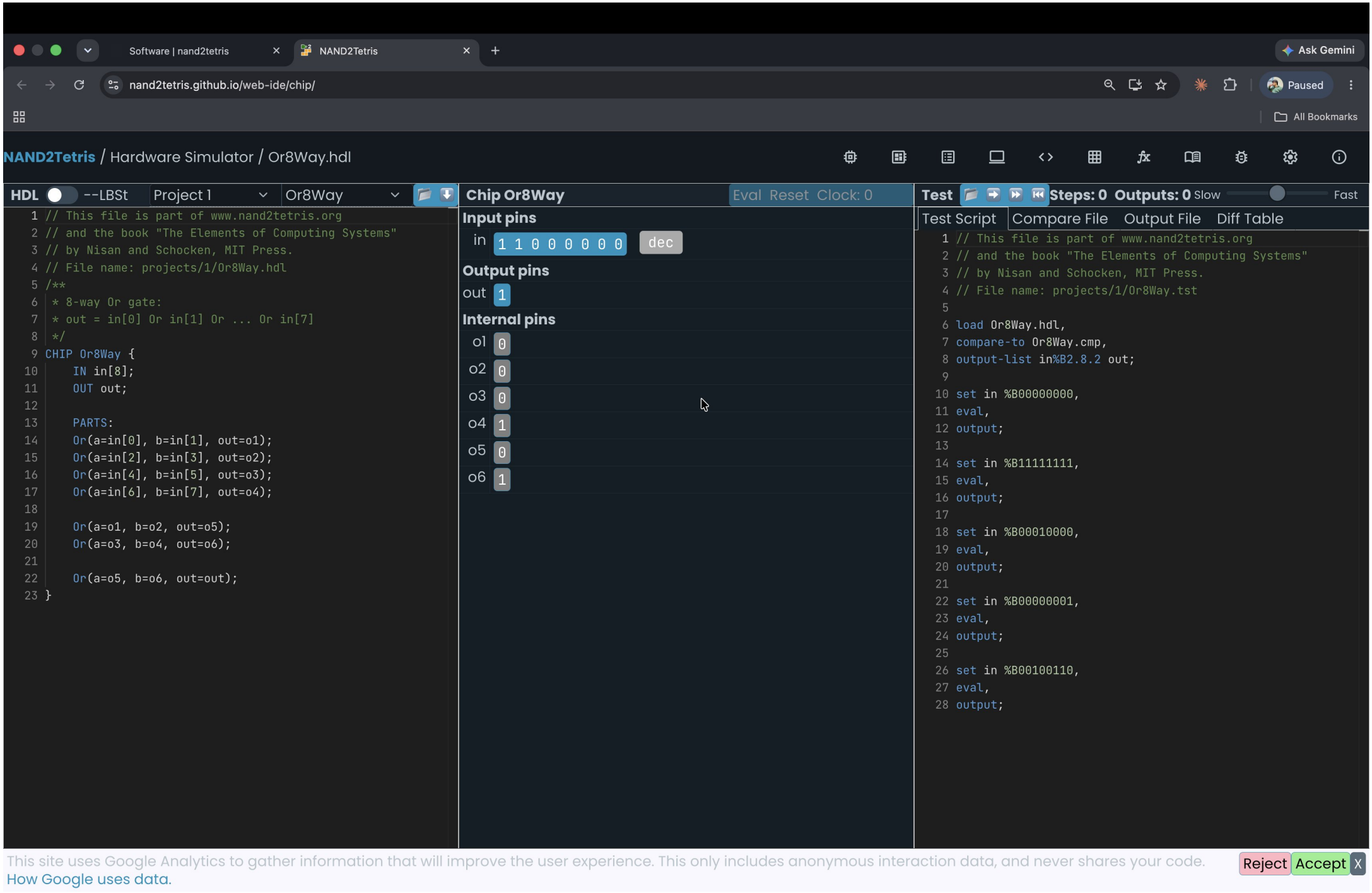

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NAND2Tetris / Hardware Simulator / DMux.hdl
HDL --LBSt Project 1 DMux
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/DMux.hdl
5 /**
6 * Demultiplexor:
7 * [a, b] = [in, 0] if sel = 0
8 * [0, in] if sel = 1
9 */
10 CHIP DMux {
11 IN in, sel;
12 OUT a, b;
13
14
15 PARTS:
16 Not(in=sel , out=notsel );
17 And(a=in , b=notsel , out=a );
18 And(a=in , b=sel , out=b );
19 }
Chip DMux
Eval Reset Clock: 0
Input pins
in 1
sel 1
Output pins
a 0
b 1
Internal pins
notsel 0
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/DMux.tst
5
6 load DMux.hdl,
7 compare-to DMux.cmp,
8 output-list in sel a b;
9
10 set in 0,
11 set sel 0,
12 eval,
13 output;
14
15 set sel 1,
16 eval,
17 output;
18
19 set in 1,
20 set sel 0,
21 eval,
22 output;
23
24 set sel 1,
25 eval,
26 output;
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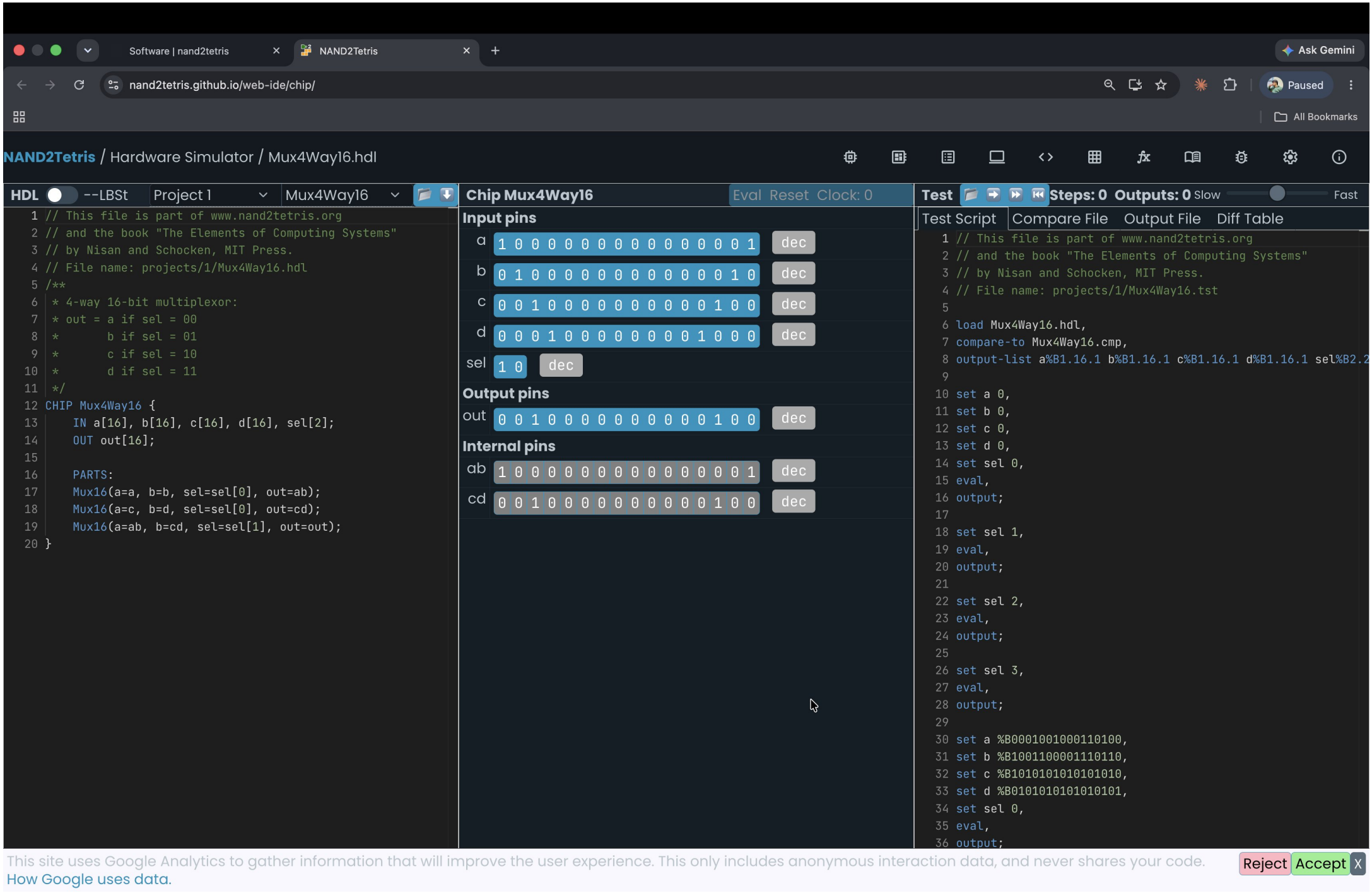

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NAND2Tetris / Hardware Simulator / Not16.hdl
HDL --LBSt Project 1 Not16
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/01/Not16.hdl
/**
* 16-bit Not gate:
* for i = 0, ..., 15:
* out[i] = Not(a[i])
*/
CHIP Not16 {
IN in[16];
OUT out[16];
PARTS:
Not(in=in[0], out=out[0]);
Not(in=in[1], out=out[1]);
Not(in=in[2], out=out[2]);
Not(in=in[3], out=out[3]);
Not(in=in[4], out=out[4]);
Not(in=in[5], out=out[5]);
Not(in=in[6], out=out[6]);
Not(in=in[7], out=out[7]);
Not(in=in[8], out=out[8]);
Not(in=in[9], out=out[9]);
Not(in=in[10], out=out[10]);
Not(in=in[11], out=out[11]);
Not(in=in[12], out=out[12]);
Not(in=in[13], out=out[13]);
Not(in=in[14], out=out[14]);
Not(in=in[15], out=out[15]);
}
Chip Not16
Eval Reset Clock: 0
Input pins
in 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1
dec
Output pins
out 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0
dec
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Not16.tst
load Not16.hdl,
compare-to Not16.cmp,
output-list in%B1.16.1 out%B1.16.1;
set in %B0000000000000000,
eval,
output;
set in %B1111111111111111,
eval,
output;
set in %B1010101010101010,
eval,
output;
set in %B0011110011000011,
eval,
output;
set in %B0001001000110100,
eval,
output;
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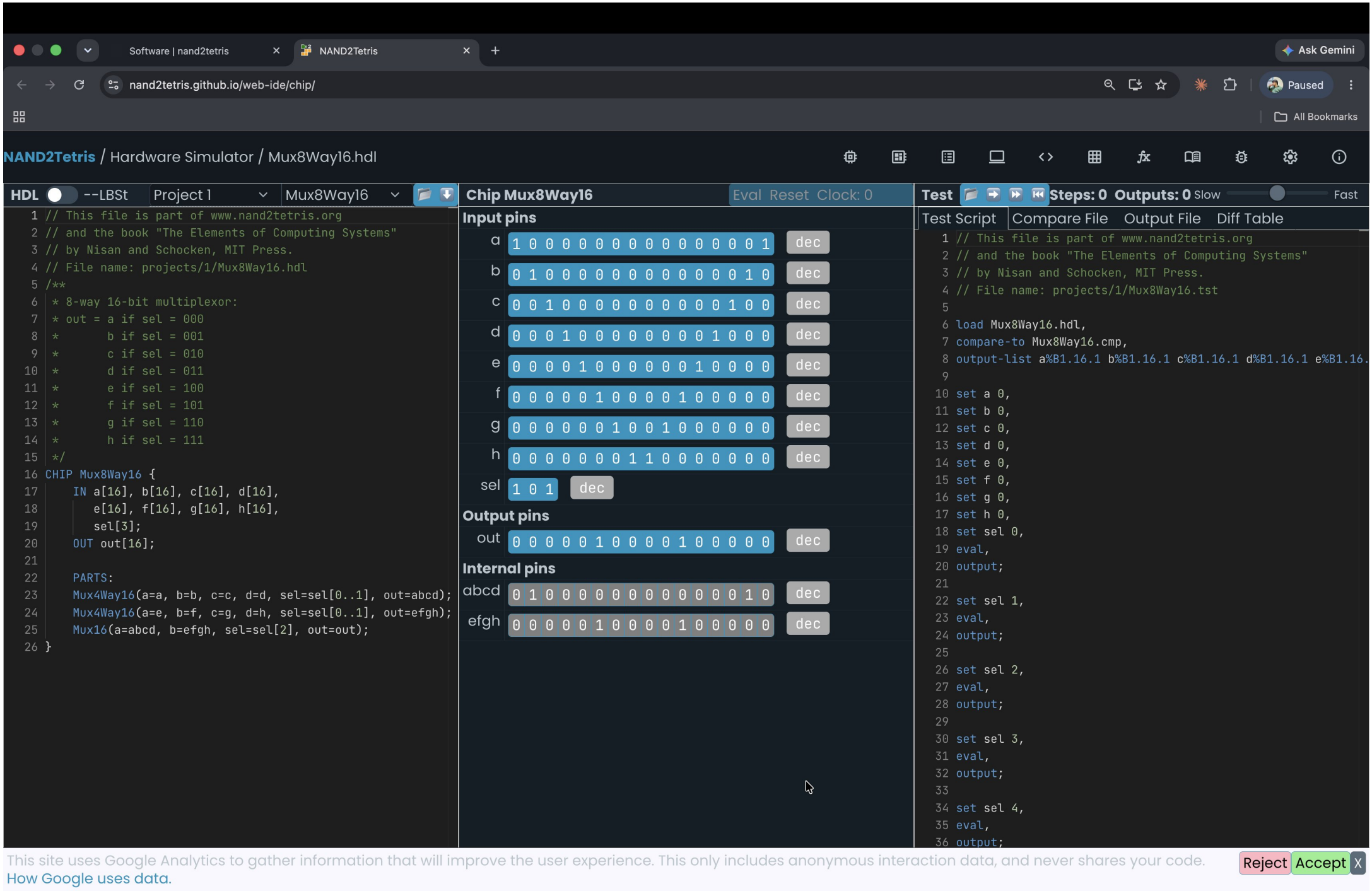

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All Bookmarks
NAND2Tetris / Hardware Simulator / And16.hdl
HDL --LBSt Project 1 And16
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/And16.hdl
/**
* 16-bit And gate:
* for i = 0, ..., 15:
* out[i] = a[i] And b[i]
*/
CHIP And16 {
IN a[16], b[16];
OUT out[16];
PARTS:
And(a=a[0], b=b[0], out=out[0]);
And(a=a[1], b=b[1], out=out[1]);
And(a=a[2], b=b[2], out=out[2]);
And(a=a[3], b=b[3], out=out[3]);
And(a=a[4], b=b[4], out=out[4]);
And(a=a[5], b=b[5], out=out[5]);
And(a=a[6], b=b[6], out=out[6]);
And(a=a[7], b=b[7], out=out[7]);
And(a=a[8], b=b[8], out=out[8]);
And(a=a[9], b=b[9], out=out[9]);
And(a=a[10], b=b[10], out=out[10]);
And(a=a[11], b=b[11], out=out[11]);
And(a=a[12], b=b[12], out=out[12]);
And(a=a[13], b=b[13], out=out[13]);
And(a=a[14], b=b[14], out=out[14]);
And(a=a[15], b=b[15], out=out[15]);
}
Chip And16
Eval Reset Clock: 0
Input pins
a 1 0 0 0 0 0 1 0 0 0 0 0 0 0 0 1 dec
b 1 0 0 0 0 0 1 0 0 0 0 0 0 0 0 1 dec
Output pins
out 1 0 0 0 0 0 1 0 0 0 0 0 0 0 0 1 dec
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/And16.tst
load And16.hdl,
compare-to And16.cmp,
output-list a%B1.16.1 b%B1.16.1 out%B1.16.1;
set a %B0000000000000000,
set b %B0000000000000000,
eval,
output;
set a %B0000000000000000,
set b %B1111111111111111,
eval,
output;
set a %B1111111111111111,
set b %B1111111111111111,
eval,
output;
set a %B1010101010101010,
set b %B0101010101010101,
eval,
output;
set a %B0011110011000011,
set b %B0000111111110000,
eval,
output;
set a %B0001001000110100,
set b %B1001100001110110,
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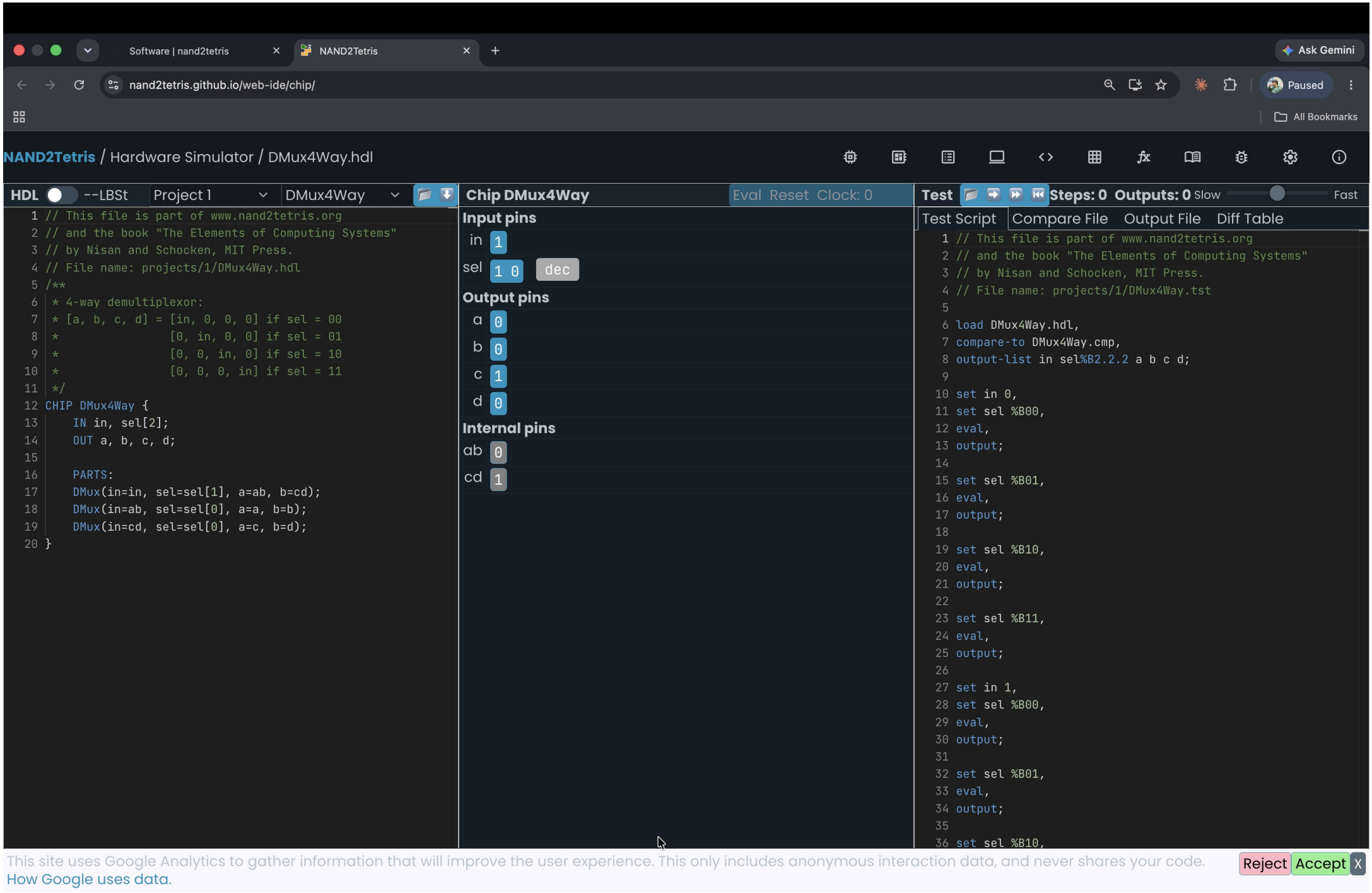

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NAND2Tetris / Hardware Simulator / Or16.hdl
HDL --LBSt Project 1 Or16
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Or16.hdl
/**
* 16-bit Or gate:
* for i = 0, ..., 15:
* out[i] = a[i] Or b[i]
*/
CHIP Or16 {
IN a[16], b[16];
OUT out[16];
PARTS:
Or(a=a[0], b=b[0], out=out[0]);
Or(a=a[1], b=b[1], out=out[1]);
Or(a=a[2], b=b[2], out=out[2]);
Or(a=a[3], b=b[3], out=out[3]);
Or(a=a[4], b=b[4], out=out[4]);
Or(a=a[5], b=b[5], out=out[5]);
Or(a=a[6], b=b[6], out=out[6]);
Or(a=a[7], b=b[7], out=out[7]);
Or(a=a[8], b=b[8], out=out[8]);
Or(a=a[9], b=b[9], out=out[9]);
Or(a=a[10], b=b[10], out=out[10]);
Or(a=a[11], b=b[11], out=out[11]);
Or(a=a[12], b=b[12], out=out[12]);
Or(a=a[13], b=b[13], out=out[13]);
Or(a=a[14], b=b[14], out=out[14]);
Or(a=a[15], b=b[15], out=out[15]);
}
Chip Or16
Eval Reset Clock: 0
Input pins
a 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
b 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 dec
Output pins
out 1 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 dec
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Or16.tst
load Or16.hdl,
compare-to Or16.cmp,
output-list a%B1.16.1 b%B1.16.1 out%B1.16.1;
set a %B0000000000000000,
set b %B0000000000000000,
eval,
output;
set a %B0000000000000000,
set b %B1111111111111111,
eval,
output;
set a %B1111111111111111,
set b %B1111111111111111,
eval,
output;
set a %B1010101010101010,
set b %B0101010101010101,
eval,
output;
set a %B0011110011000011,
set b %B0000111111110000,
eval,
output;
set a %B0001001000110100,
set b %B1001100001110110,
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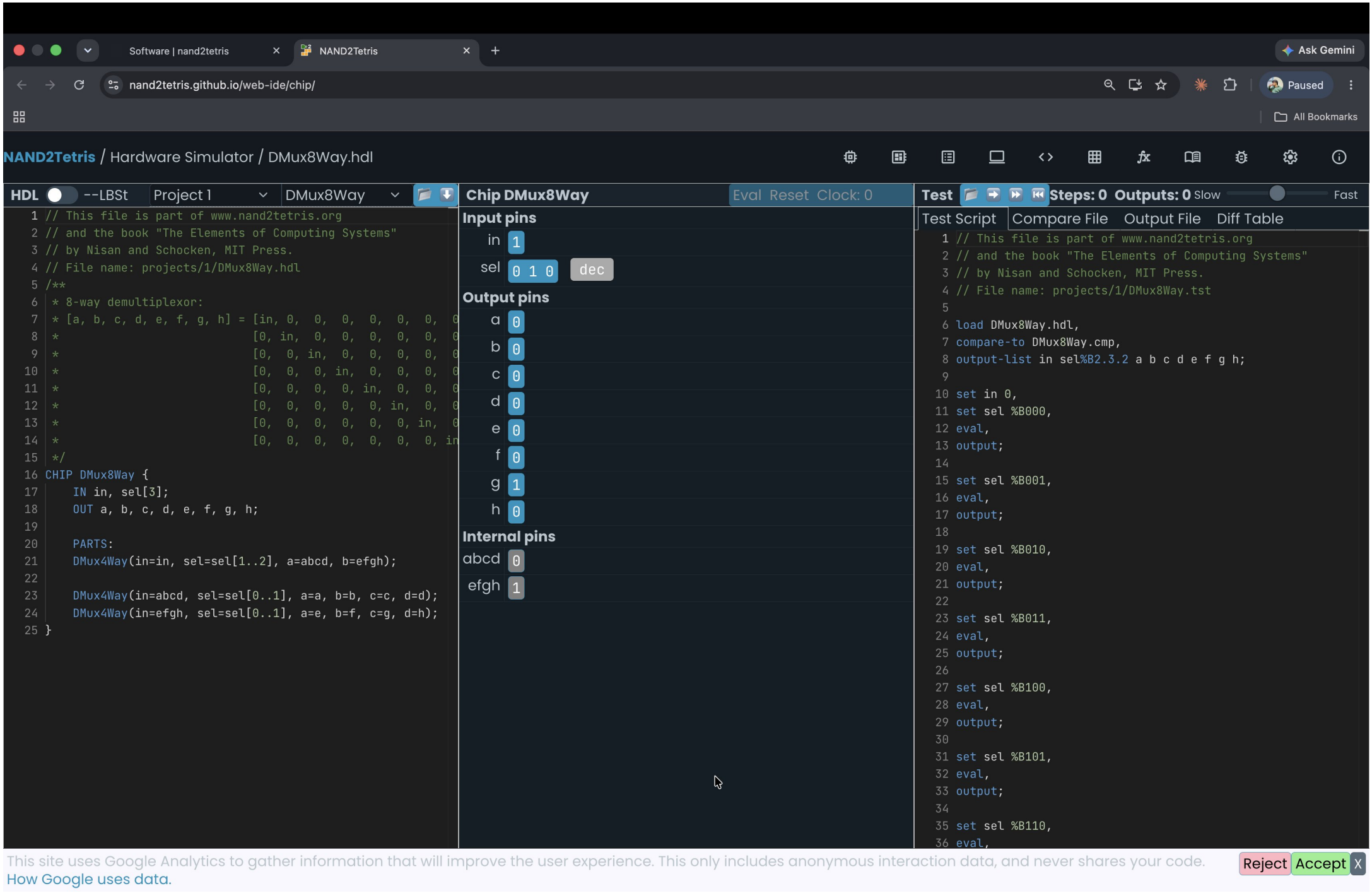

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All Bookmarks
NAND2Tetris / Hardware Simulator / Mux16.hdl
HDL --LBSt Project 1 Mux16
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Mux16.hdl
/**
* 16-bit multiplexor:
* for i = 0, ..., 15:
* if (sel = 0) out[i] = a[i], else out[i] = b[i]
*/
CHIP Mux16 {
IN a[16], b[16], sel;
OUT out[16];
PARTS:
Mux(a=a[0], b=b[0], sel=sel, out=out[0]);
Mux(a=a[1], b=b[1], sel=sel, out=out[1]);
Mux(a=a[2], b=b[2], sel=sel, out=out[2]);
Mux(a=a[3], b=b[3], sel=sel, out=out[3]);
Mux(a=a[4], b=b[4], sel=sel, out=out[4]);
Mux(a=a[5], b=b[5], sel=sel, out=out[5]);
Mux(a=a[6], b=b[6], sel=sel, out=out[6]);
Mux(a=a[7], b=b[7], sel=sel, out=out[7]);
Mux(a=a[8], b=b[8], sel=sel, out=out[8]);
Mux(a=a[9], b=b[9], sel=sel, out=out[9]);
Mux(a=a[10], b=b[10], sel=sel, out=out[10]);
Mux(a=a[11], b=b[11], sel=sel, out=out[11]);
Mux(a=a[12], b=b[12], sel=sel, out=out[12]);
Mux(a=a[13], b=b[13], sel=sel, out=out[13]);
Mux(a=a[14], b=b[14], sel=sel, out=out[14]);
Mux(a=a[15], b=b[15], sel=sel, out=out[15]);
}
Chip Mux16
Eval Reset Clock: 0
Input pins
a 1 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 dec
b 0 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0 dec
sel 1
Output pins
out 0 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0 dec
Test Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Mux16.tst
load Mux16.hdl,
compare-to Mux16.cmp,
output-list a%B1.16.1 b%B1.16.1 sel out%B1.16.1;
set a 0,
set b 0,
set sel 0,
eval,
output;
set sel 1,
eval,
output;
set a %B0000000000000000,
set b %B0001001000110100,
set sel 0,
eval,
output;
set sel 1,
eval,
output;
set a %B1001100001110110,
set b %B0000000000000000,
set sel 0,
eval,
output;
set sel 1,
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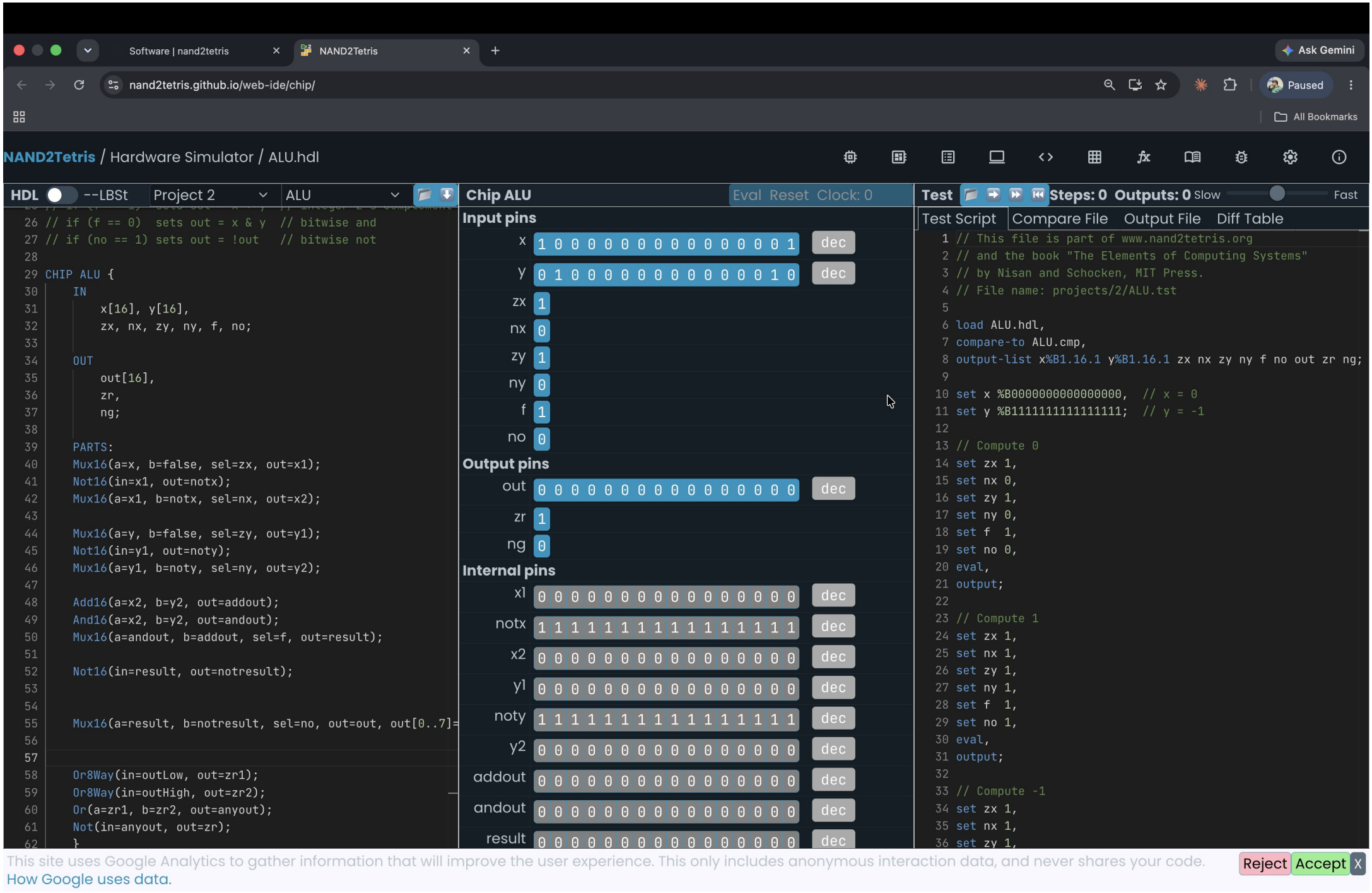

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NAND2Tetris / Hardware Simulator / Or8Way.hdl
HDL --LBSt Project 1 Or8Way
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Or8Way.hdl
/**
* 8-way Or gate:
* out = in[0] Or in[1] Or ... Or in[7]
*/
CHIP Or8Way {
IN in[8];
OUT out;
PARTS:
Or(a=in[0], b=in[1], out=o1);
Or(a=in[2], b=in[3], out=o2);
Or(a=in[4], b=in[5], out=o3);
Or(a=in[6], b=in[7], out=o4);
Or(a=o1, b=o2, out=o5);
Or(a=o3, b=o4, out=o6);
Or(a=o5, b=o6, out=out);
}
Chip Or8Way
Eval Reset Clock: 0
Input pins
in 1 1 0 0 0 0 0 0
dec
Output pins
out 1
Internal pins
o1 0
o2 0
o3 0
o4 1
o5 0
o6 1
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Or8Way.tst
load Or8Way.hdl,
compare-to Or8Way.cmp,
output-list in%B2.8.2 out;
set in %B00000000,
eval,
output;
set in %B11111111,
eval,
output;
set in %B00010000,
eval,
output;
set in %B00000001,
eval,
output;
set in %B00100110,
eval,
output;
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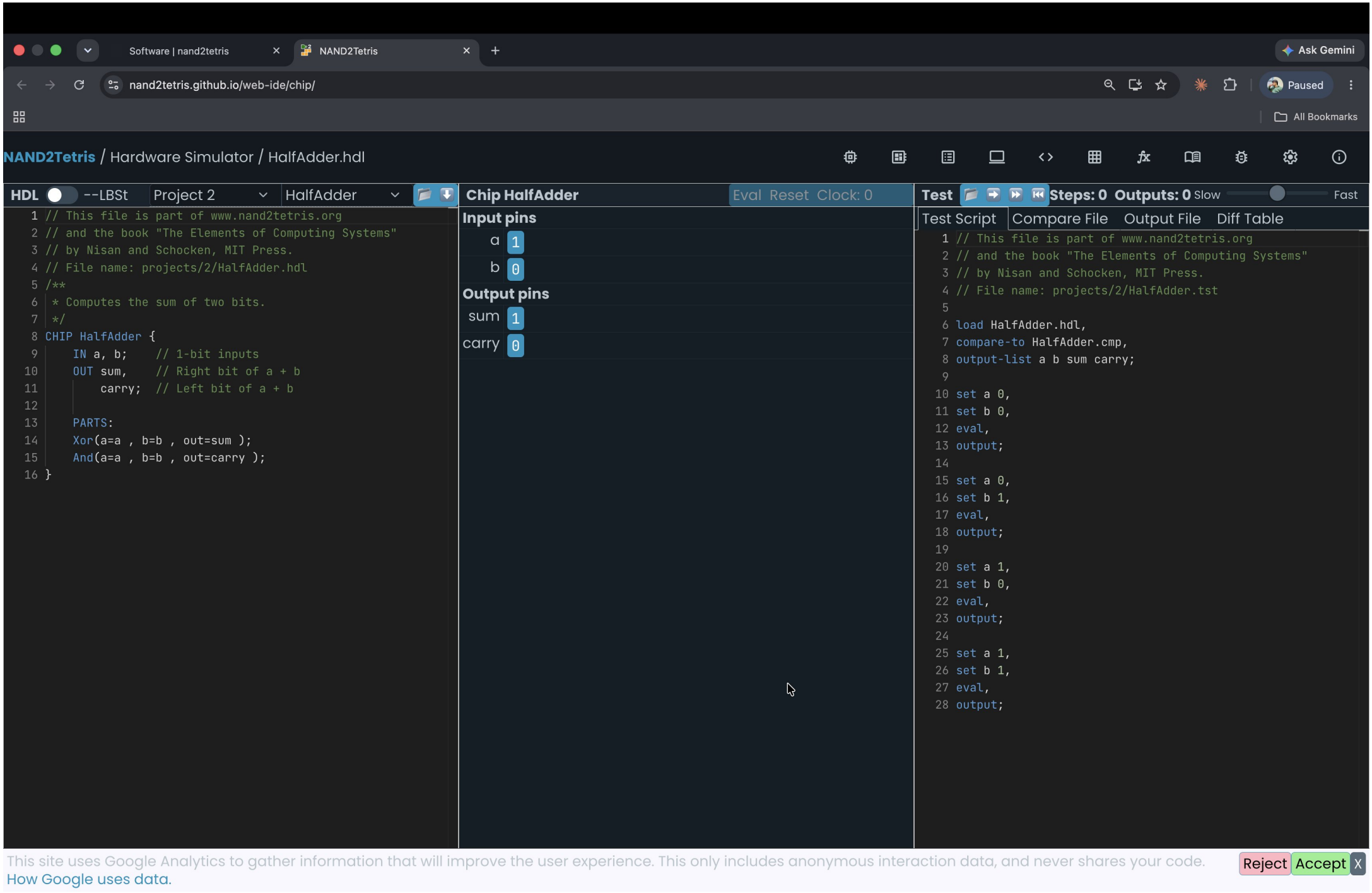

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NAND2Tetris / Hardware Simulator / Mux4Way16.hdl
HDL --LBSt Project 1 Mux4Way16
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Mux4Way16.hdl
/**
* 4-way 16-bit multiplexor:
* out = a if sel = 00
*       b if sel = 01
*       c if sel = 10
*       d if sel = 11
*/
CHIP Mux4Way16 {
IN a[16], b[16], c[16], d[16], sel[2];
OUT out[16];
PARTS:
Mux16(a=a, b=b, sel=sel[0], out=ab);
Mux16(a=c, b=d, sel=sel[0], out=cd);
Mux16(a=ab, b=cd, sel=sel[1], out=out);
}
Chip Mux4Way16
Eval Reset Clock: 0
Input pins
a 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
b 0 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0 dec
c 0 0 1 0 0 0 0 0 0 0 0 0 0 1 0 0 dec
d 0 0 0 1 0 0 0 0 0 0 0 0 1 0 0 0 dec
sel 1 0 dec
Output pins
out 0 0 1 0 0 0 0 0 0 0 0 0 0 1 0 0 dec
Internal pins
ab 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
cd 0 0 1 0 0 0 0 0 0 0 0 0 0 1 0 0 dec
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Mux4Way16.tst
load Mux4Way16.hdl,
compare-to Mux4Way16.cmp,
output-list a%B1.16.1 b%B1.16.1 c%B1.16.1 d%B1.16.1 sel%B2.2
set a 0,
set b 0,
set c 0,
set d 0,
set sel 0,
eval,
output;
set sel 1,
eval,
output;
set sel 2,
eval,
output;
set sel 3,
eval,
output;
set a %B0001001000110100,
set b %B1001100001110110,
set c %B1010101010101010,
set d %B0101010101010101,
set sel 0,
eval,
output;
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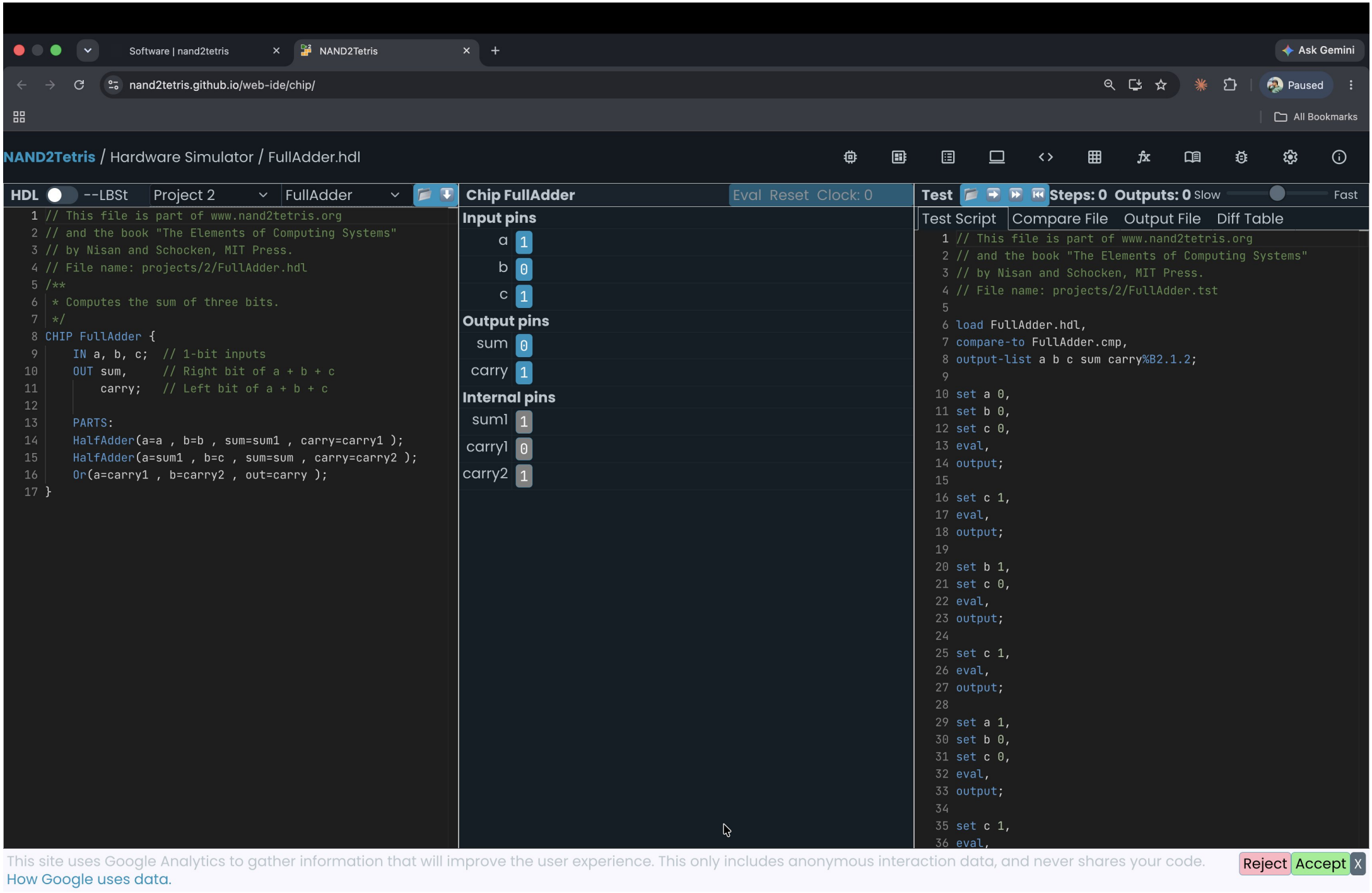

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NAND2Tetris / Hardware Simulator / Mux8Way16.hdl
HDL --LBSt Project 1 Mux8Way16
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Mux8Way16.hdl
/**
* 8-way 16-bit multiplexor:
* out = a if sel = 000
*       b if sel = 001
*       c if sel = 010
*       d if sel = 011
*       e if sel = 100
*       f if sel = 101
*       g if sel = 110
*       h if sel = 111
*/
CHIP Mux8Way16 {
IN a[16], b[16], c[16], d[16],
e[16], f[16], g[16], h[16],
sel[3];
OUT out[16];
PARTS:
Mux4Way16(a=a, b=b, c=c, d=d, sel=sel[0..1], out=abcd);
Mux4Way16(a=e, b=f, c=g, d=h, sel=sel[0..1], out=efgh);
Mux16(a=abcd, b=efgh, sel=sel[2], out=out);
}
Chip Mux8Way16
Eval Reset Clock: 0
Input pins
a 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
b 0 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0 dec
c 0 0 1 0 0 0 0 0 0 0 0 0 0 1 0 0 dec
d 0 0 0 1 0 0 0 0 0 0 0 0 1 0 0 0 dec
e 0 0 0 0 1 0 0 0 0 0 0 1 0 0 0 0 dec
f 0 0 0 0 0 1 0 0 0 0 1 0 0 0 0 0 dec
g 0 0 0 0 0 0 1 0 0 1 0 0 0 0 0 0 dec
h 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 dec
sel 1 0 1 dec
Output pins
out 0 0 0 0 0 1 0 0 0 0 1 0 0 0 0 0 dec
Internal pins
abcd 0 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0 dec
efgh 0 0 0 0 0 1 0 0 0 0 1 0 0 0 0 0 dec
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Mux8Way16.tst
load Mux8Way16.hdl,
compare-to Mux8Way16.cmp,
output-list a%B1.16.1 b%B1.16.1 c%B1.16.1 d%B1.16.1 e%B1.16.
set a 0,
set b 0,
set c 0,
set d 0,
set e 0,
set f 0,
set g 0,
set h 0,
set sel 0,
eval,
output;
set sel 1,
eval,
output;
set sel 2,
eval,
output;
set sel 3,
eval,
output;
set sel 4,
eval,
output;
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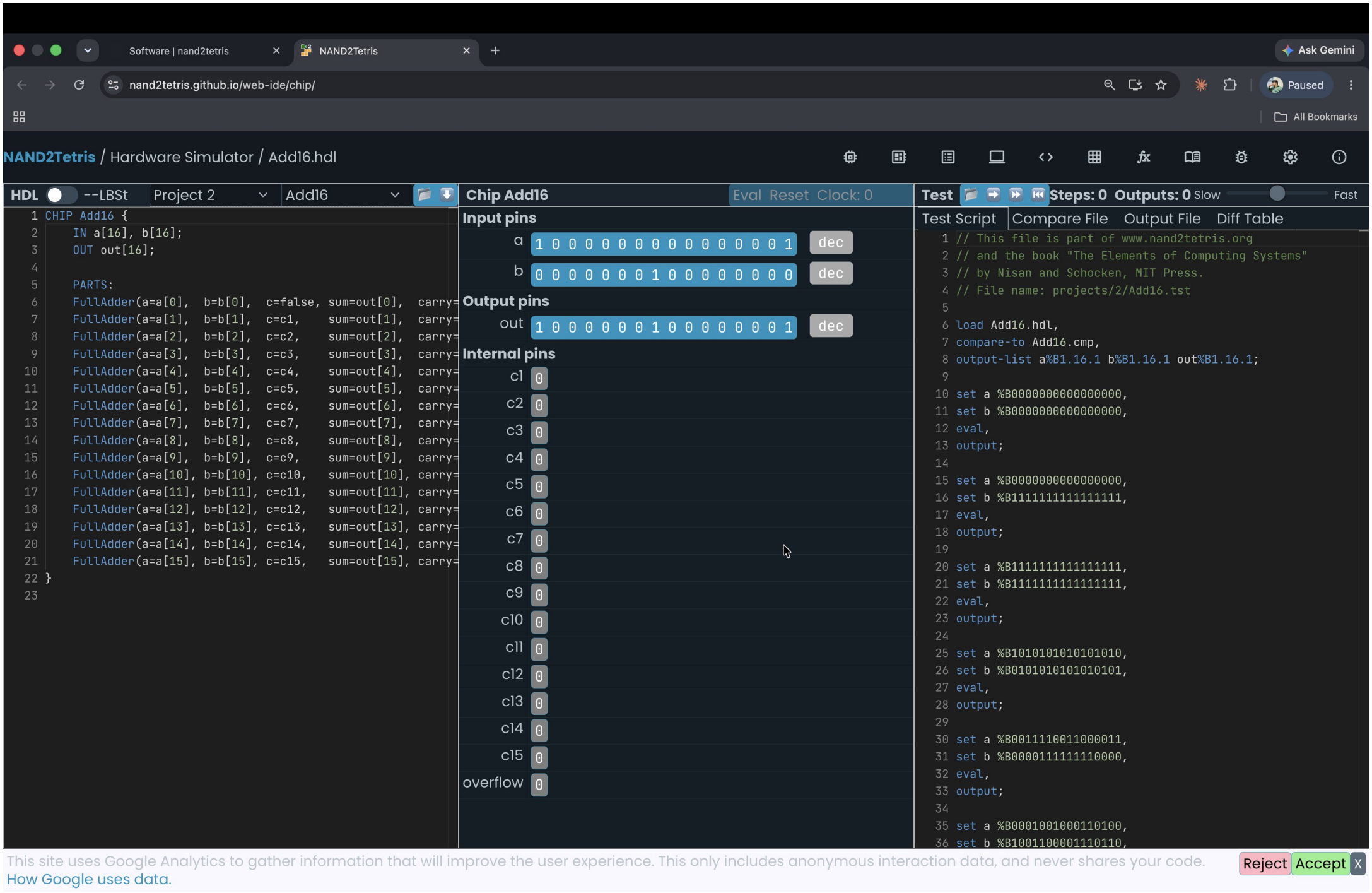

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NAND2Tetris / Hardware Simulator / DMux4Way.hdl
HDL --LBSt Project 1 DMux4Way
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/DMux4Way.hdl
/**
* 4-way demultiplexor:
* [a, b, c, d] = [in, 0, 0, 0] if sel = 00
* [0, in, 0, 0] if sel = 01
* [0, 0, in, 0] if sel = 10
* [0, 0, 0, in] if sel = 11
*/
CHIP DMux4Way {
IN in, sel[2];
OUT a, b, c, d;
PARTS:
DMux(in=in, sel=sel[1], a=ab, b=cd);
DMux(in=ab, sel=sel[0], a=a, b=b);
DMux(in=cd, sel=sel[0], a=c, b=d);
}
Chip DMux4Way
Eval Reset Clock: 0
Input pins
in 1
sel 1 0 dec
Output pins
a 0
b 0
c 1
d 0
Internal pins
ab 0
cd 1
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/DMux4Way.tst
load DMux4Way.hdl,
compare-to DMux4Way.cmp,
output-list in sel%B2.2.2 a b c d;
set in 0,
set sel %B00,
eval,
output;
set sel %B01,
eval,
output;
set sel %B10,
eval,
output;
set sel %B11,
eval,
output;
set in 1,
set sel %B00,
eval,
output;
set sel %B01,
eval,
output;
set sel %B10,
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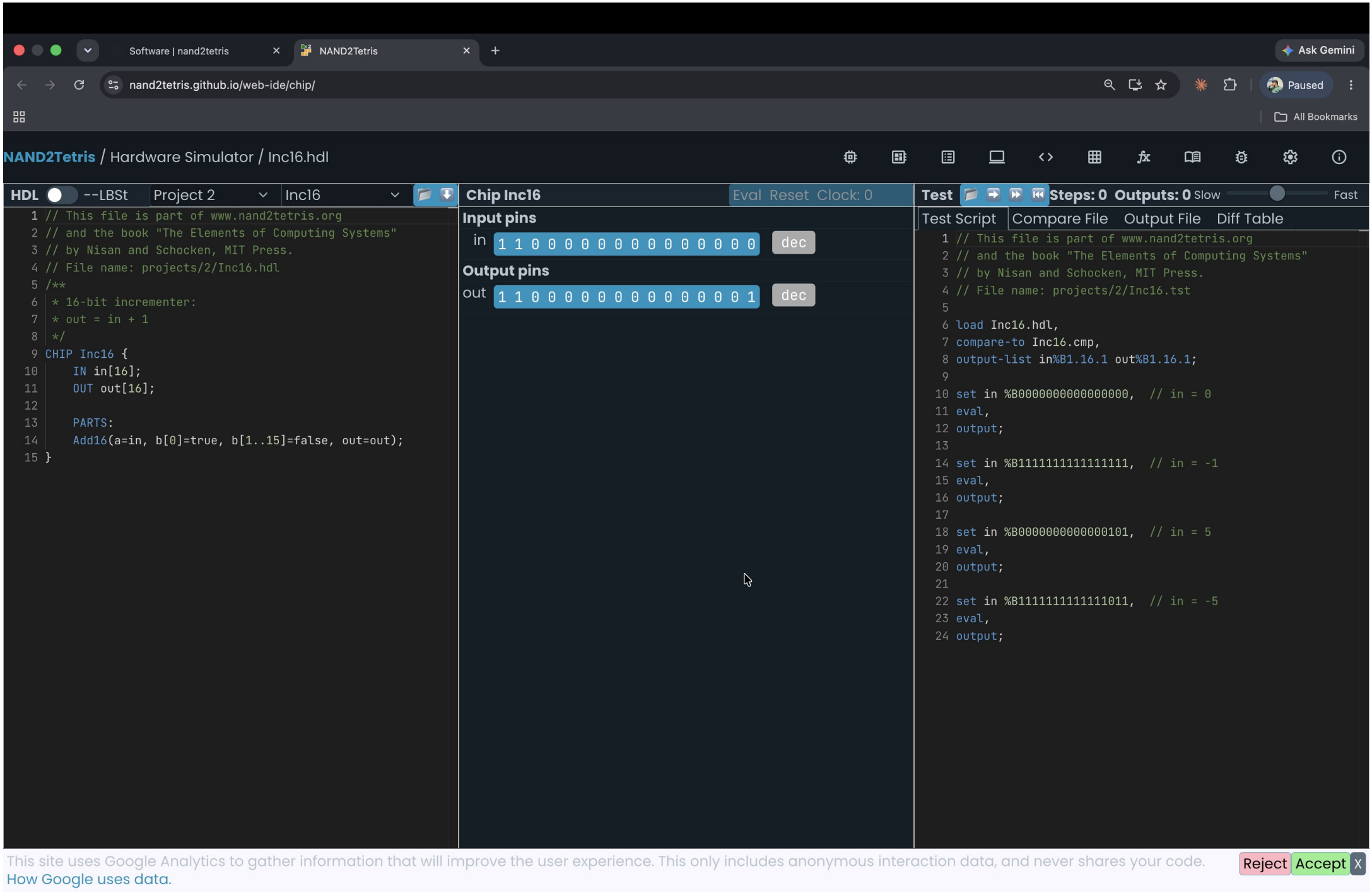

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NAND2Tetris / Hardware Simulator / DMux8Way.hdl
HDL --LBSt Project 1 DMux8Way
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/DMux8Way.hdl
/**
* 8-way demultiplexor:
* [a, b, c, d, e, f, g, h] = [in, 0, 0, 0, 0, 0, 0, 0
* [0, in, 0, 0, 0, 0, 0, 0
* [0, 0, in, 0, 0, 0, 0, 0
* [0, 0, 0, in, 0, 0, 0, 0
* [0, 0, 0, 0, in, 0, 0, 0
* [0, 0, 0, 0, 0, in, 0, 0
* [0, 0, 0, 0, 0, 0, in, 0
* [0, 0, 0, 0, 0, 0, 0, in
*/
CHIP DMux8Way {
IN in, sel[3];
OUT a, b, c, d, e, f, g, h;
PARTS:
DMux4Way(in=in, sel=sel[1..2], a=abcd, b=efgh);
DMux4Way(in=abcd, sel=sel[0..1], a=a, b=b, c=c, d=d);
DMux4Way(in=efgh, sel=sel[0..1], a=e, b=f, c=g, d=h);
}
Chip DMux8Way
Eval Reset Clock: 0
Input pins
in 1
sel 0 1 0 dec
Output pins
a 0
b 0
c 0
d 0
e 0
f 0
g 1
h 0
Internal pins
abcd 0
efgh 1
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/DMux8Way.tst
load DMux8Way.hdl,
compare-to DMux8Way.cmp,
output-list in sel%B2.3.2 a b c d e f g h;
set in 0,
set sel %B000,
eval,
output;
set sel %B001,
eval,
output;
set sel %B010,
eval,
output;
set sel %B011,
eval,
output;
set sel %B100,
eval,
output;
set sel %B101,
eval,
output;
set sel %B110,
eval,
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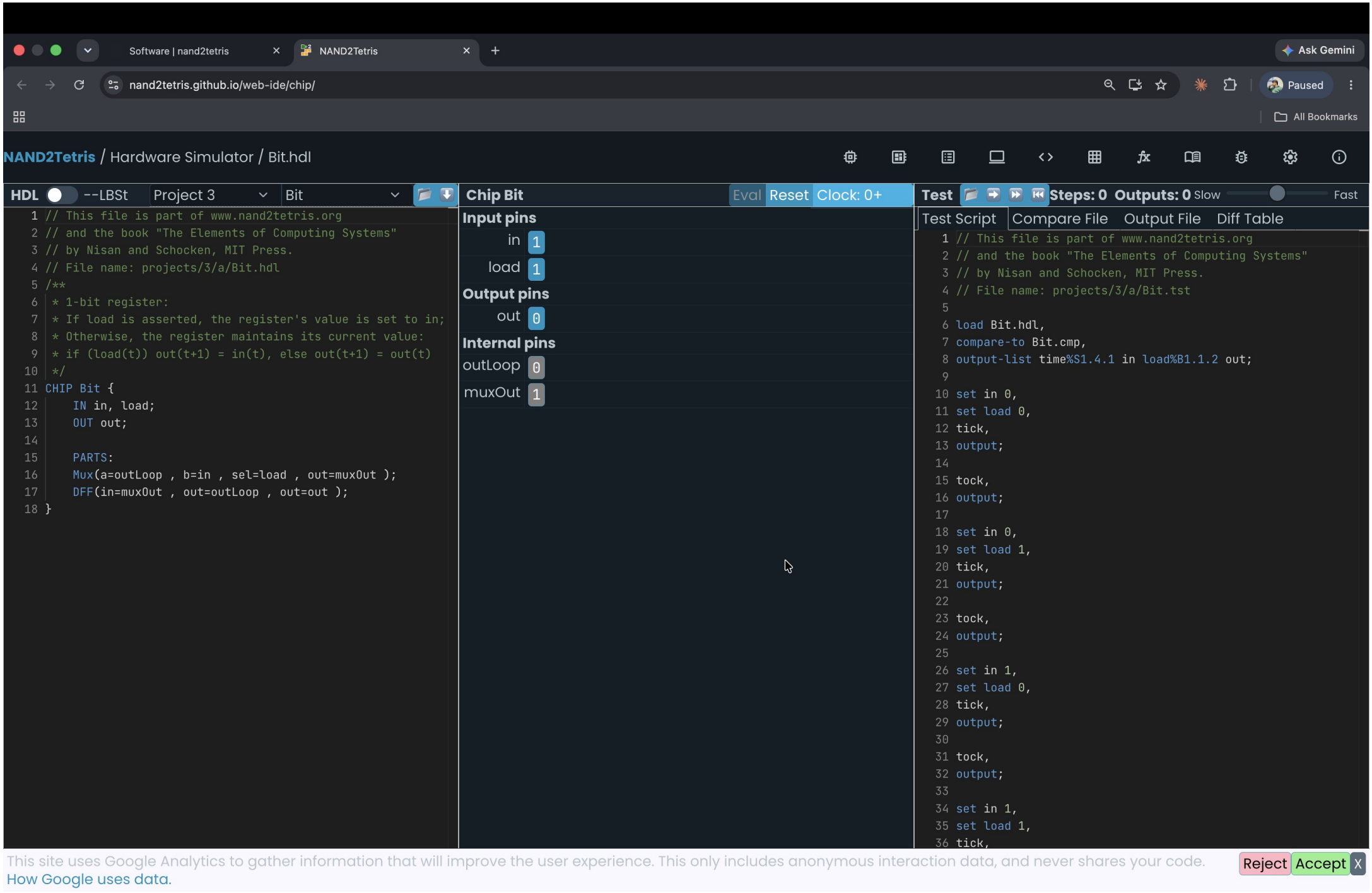

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NAND2Tetris / Hardware Simulator / ALU.hdl
HDL --LBSt Project 2 ALU
// if (f == 0) sets out = x & y // bitwise and
// if (no == 1) sets out = !out // bitwise not
CHIP ALU {
IN
x[16], y[16],
zx, nx, zy, ny, f, no;
OUT
out[16],
zr,
ng;
PARTS:
Mux16(a=x, b=false, sel=zx, out=x1);
Not16(in=x1, out=notx);
Mux16(a=x1, b=notx, sel=nx, out=x2);
Mux16(a=y, b=false, sel=zy, out=y1);
Not16(in=y1, out=noty);
Mux16(a=y1, b=noty, sel=ny, out=y2);
Add16(a=x2, b=y2, out=addout);
And16(a=x2, b=y2, out=andout);
Mux16(a=andout, b=addout, sel=f, out=result);
Not16(in=result, out=notresult);
Mux16(a=result, b=notresult, sel=no, out=out, out[0..7]=
Or8Way(in=outLow, out=zr1);
Or8Way(in=outHigh, out=zr2);
Or(a=zr1, b=zr2, out=anyout);
Not(in=anyout, out=zr);
}
Chip ALU
Eval Reset Clock: 0
Input pins
x 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
y 0 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0 dec
zx 1
nx 0
zy 1
ny 0
f 1
no 0
Output pins
out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
zr 1
ng 0
Internal pins
x1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
notx 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 dec
x2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
y1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
noty 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 dec
y2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
addout 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
andout 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
result 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
Test Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/ALU.tst
load ALU.hdl,
compare-to ALU.cmp,
output-list x%B1.16.1 y%B1.16.1 zx nx zy ny f no out zr ng;
set x %B0000000000000000, // x = 0
set y %B1111111111111111; // y = -1
// Compute 0
set zx 1,
set nx 0,
set zy 1,
set ny 0,
set f 1,
set no 0,
eval,
output;
// Compute 1
set zx 1,
set nx 1,
set zy 1,
set ny 1,
set f 1,
set no 1,
eval,
output;
// Compute -1
set zx 1,
set nx 1,
set zy 1,
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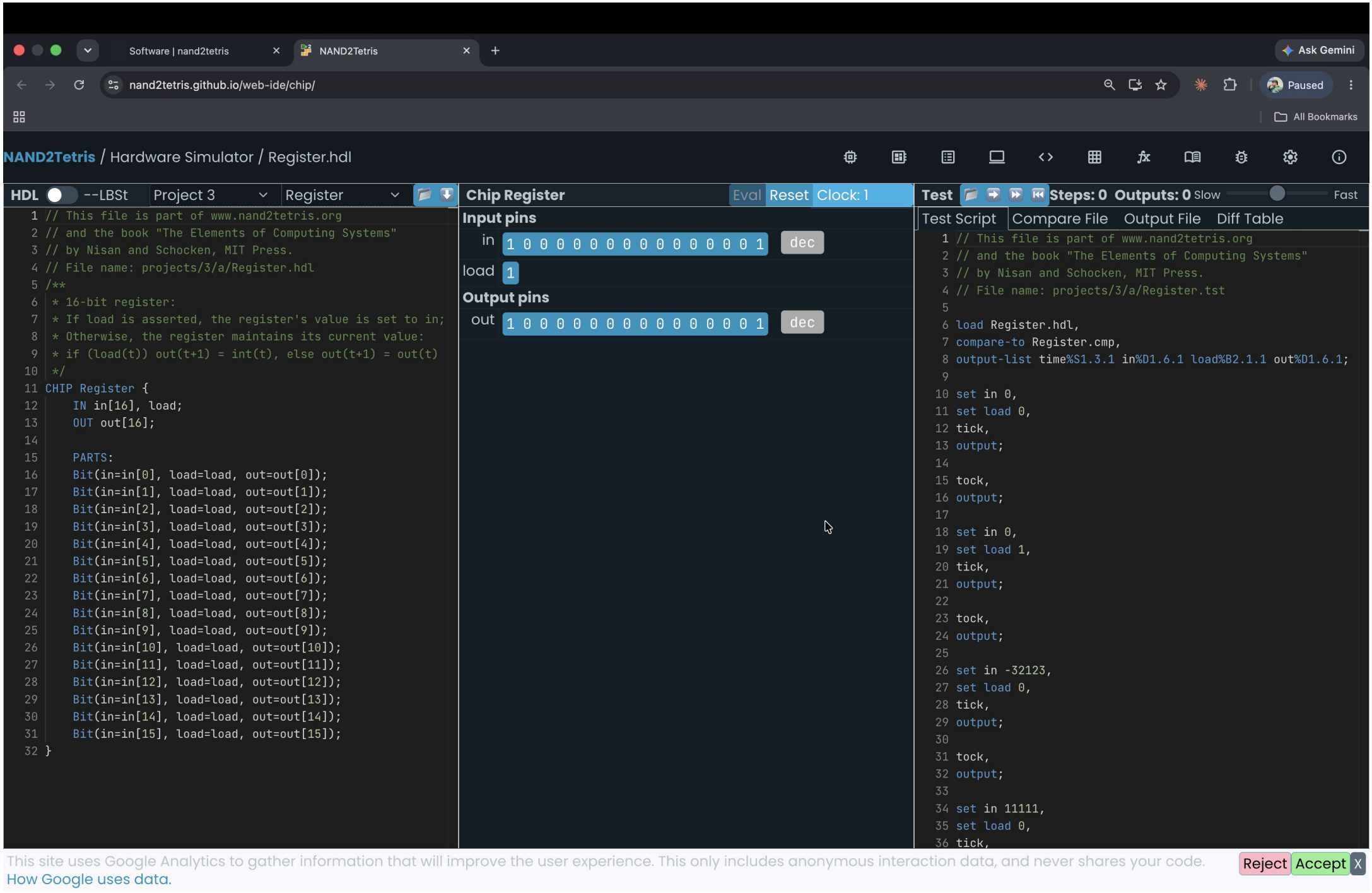

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NAND2Tetris / Hardware Simulator / HalfAdder.hdl
HDL --LBSt Project 2 HalfAdder
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/HalfAdder.hdl
/**
* Computes the sum of two bits.
*/
CHIP HalfAdder {
IN a, b; // 1-bit inputs
OUT sum, // Right bit of a + b
carry; // Left bit of a + b
PARTS:
Xor(a=a , b=b , out=sum );
And(a=a , b=b , out=carry );
}
Chip HalfAdder Eval Reset Clock: 0
Input pins
a 1
b 0
Output pins
sum 1
carry 0
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/HalfAdder.tst
load HalfAdder.hdl,
compare-to HalfAdder.cmp,
output-list a b sum carry;
set a 0,
set b 0,
eval,
output;
set a 0,
set b 1,
eval,
output;
set a 1,
set b 0,
eval,
output;
set a 1,
set b 1,
eval,
output;
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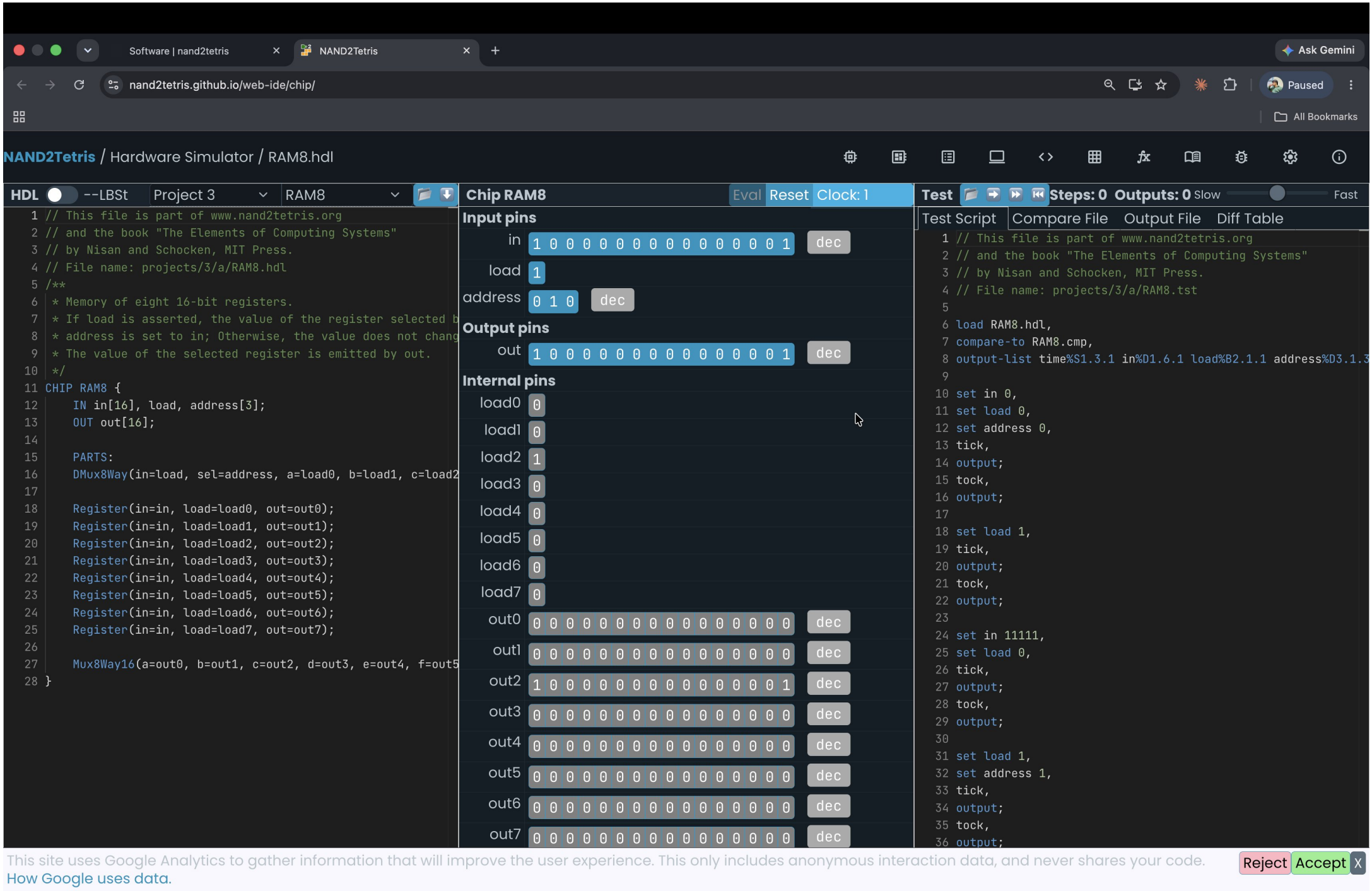

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NAND2Tetris / Hardware Simulator / FullAdder.hdl
HDL --LBSt Project 2 FullAdder
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/FullAdder.hdl
/**
* Computes the sum of three bits.
*/
CHIP FullAdder {
IN a, b, c; // 1-bit inputs
OUT sum, // Right bit of a + b + c
carry; // Left bit of a + b + c
PARTS:
HalfAdder(a=a , b=b , sum=sum1 , carry=carry1 );
HalfAdder(a=sum1 , b=c , sum=sum , carry=carry2 );
Or(a=carry1 , b=carry2 , out=carry );
}
Chip FullAdder
Eval Reset Clock: 0
Input pins
a 1
b 0
c 1
Output pins
sum 0
carry 1
Internal pins
sum1 1
carry1 0
carry2 1
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/FullAdder.tst
load FullAdder.hdl,
compare-to FullAdder.cmp,
output-list a b c sum carry%B2.1.2;
set a 0,
set b 0,
set c 0,
eval,
output;
set c 1,
eval,
output;
set b 1,
set c 0,
eval,
output;
set c 1,
eval,
output;
set a 1,
set b 0,
set c 0,
eval,
output;
set c 1,
eval,
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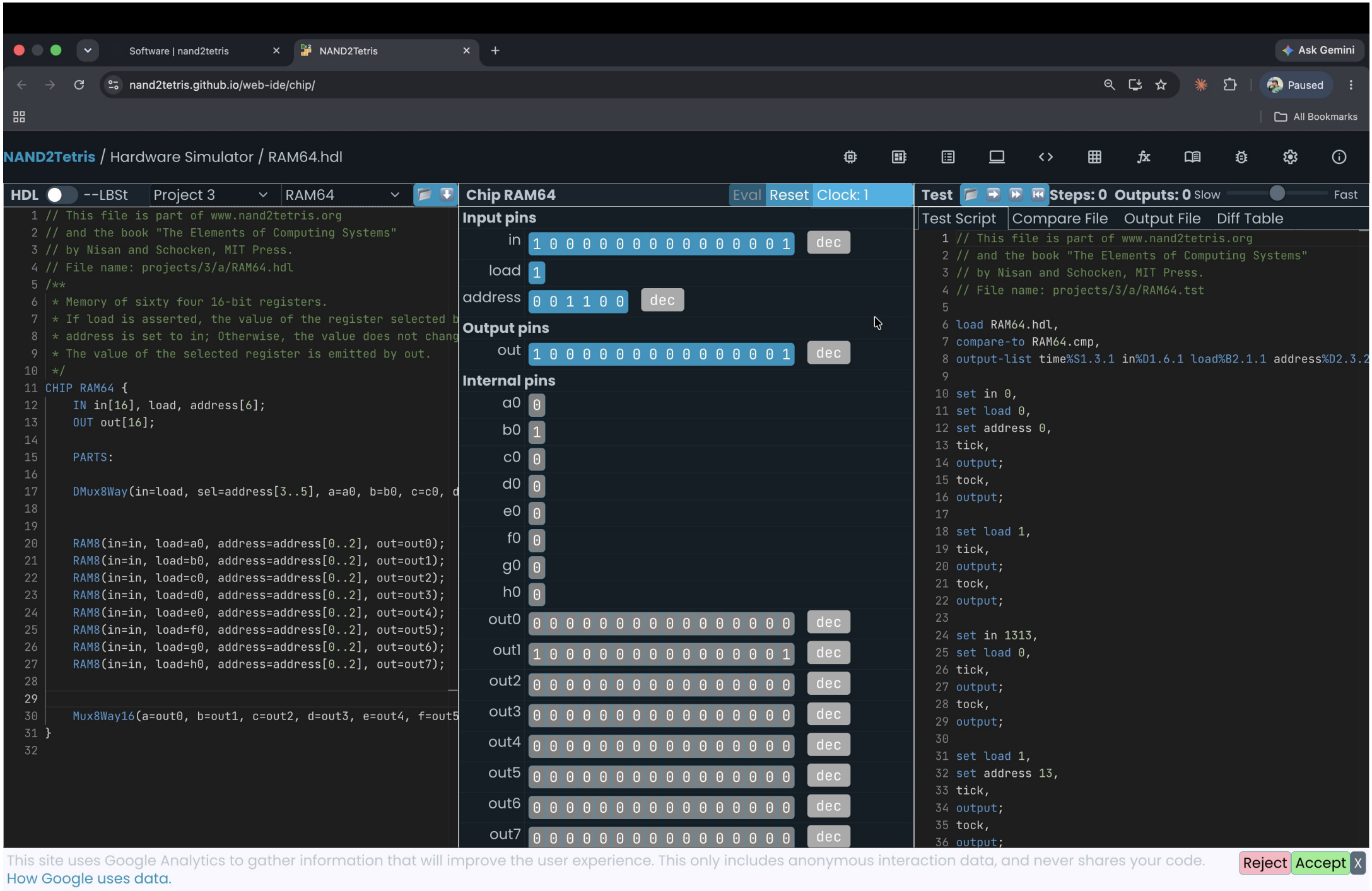

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NAND2Tetris / Hardware Simulator / Add16.hdl
HDL --LBSt Project 2 Add16
CHIP Add16 {
IN a[16], b[16];
OUT out[16];
PARTS:
FullAdder(a=a[0], b=b[0], c=false, sum=out[0], carry=
FullAdder(a=a[1], b=b[1], c=c1, sum=out[1], carry=
FullAdder(a=a[2], b=b[2], c=c2, sum=out[2], carry=
FullAdder(a=a[3], b=b[3], c=c3, sum=out[3], carry=
FullAdder(a=a[4], b=b[4], c=c4, sum=out[4], carry=
FullAdder(a=a[5], b=b[5], c=c5, sum=out[5], carry=
FullAdder(a=a[6], b=b[6], c=c6, sum=out[6], carry=
FullAdder(a=a[7], b=b[7], c=c7, sum=out[7], carry=
FullAdder(a=a[8], b=b[8], c=c8, sum=out[8], carry=
FullAdder(a=a[9], b=b[9], c=c9, sum=out[9], carry=
FullAdder(a=a[10], b=b[10], c=c10, sum=out[10], carry=
FullAdder(a=a[11], b=b[11], c=c11, sum=out[11], carry=
FullAdder(a=a[12], b=b[12], c=c12, sum=out[12], carry=
FullAdder(a=a[13], b=b[13], c=c13, sum=out[13], carry=
FullAdder(a=a[14], b=b[14], c=c14, sum=out[14], carry=
FullAdder(a=a[15], b=b[15], c=c15, sum=out[15], carry=
}
Chip Add16 Eval Reset Clock: 0
Input pins
a 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
b 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 dec
Output pins
out 1 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 dec
Internal pins
c1 0
c2 0
c3 0
c4 0
c5 0
c6 0
c7 0
c8 0
c9 0
c10 0
c11 0
c12 0
c13 0
c14 0
c15 0
overflow 0
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/Add16.tst
load Add16.hdl,
compare-to Add16.cmp,
output-list a%B1.16.1 b%B1.16.1 out%B1.16.1;
set a %B0000000000000000,
set b %B0000000000000000,
eval,
output;
set a %B0000000000000000,
set b %B1111111111111111,
eval,
output;
set a %B1111111111111111,
set b %B1111111111111111,
eval,
output;
set a %B1010101010101010,
set b %B0101010101010101,
eval,
output;
set a %B0011110011000011,
set b %B0000111111110000,
eval,
output;
set a %B0001001000110100,
set b %B1001100001110110,
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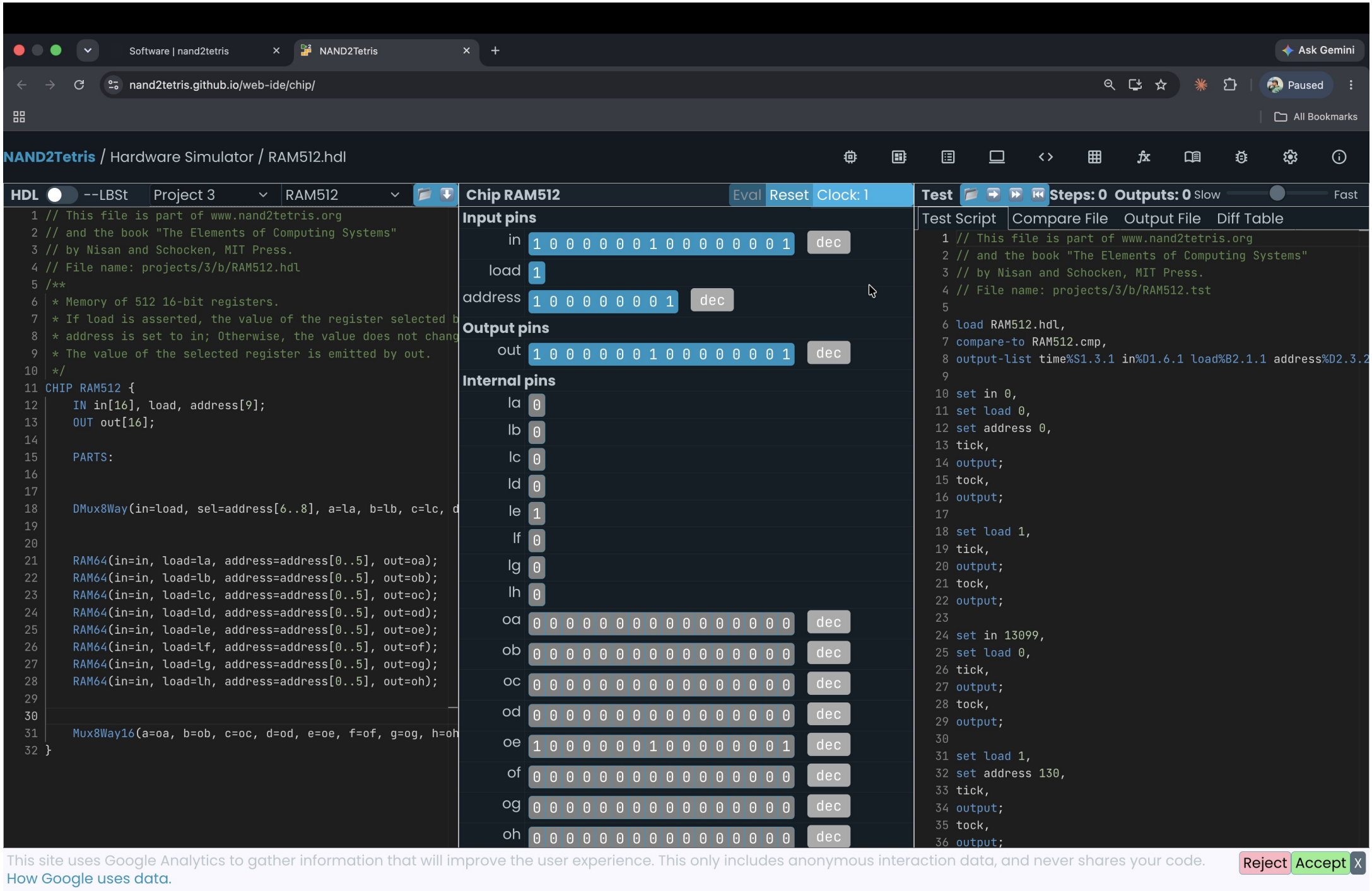

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NAND2Tetris / Hardware Simulator / Inc16.hdl
HDL --LBSt Project 2 Inc16
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/Inc16.hdl
/**
* 16-bit incrementer:
* out = in + 1
*/
CHIP Inc16 {
IN in[16];
OUT out[16];
PARTS:
Add16(a=in, b[0]=true, b[1..15]=false, out=out);
}
Chip Inc16 Eval Reset Clock: 0
Input pins
in 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
Output pins
out 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
Test Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/Inc16.tst
load Inc16.hdl,
compare-to Inc16.cmp,
output-list in%B1.16.1 out%B1.16.1;
set in %B0000000000000000, // in = 0
eval,
output;
set in %B1111111111111111, // in = -1
eval,
output;
set in %B0000000000000101, // in = 5
eval,
output;
set in %B1111111111111011, // in = -5
eval,
output;
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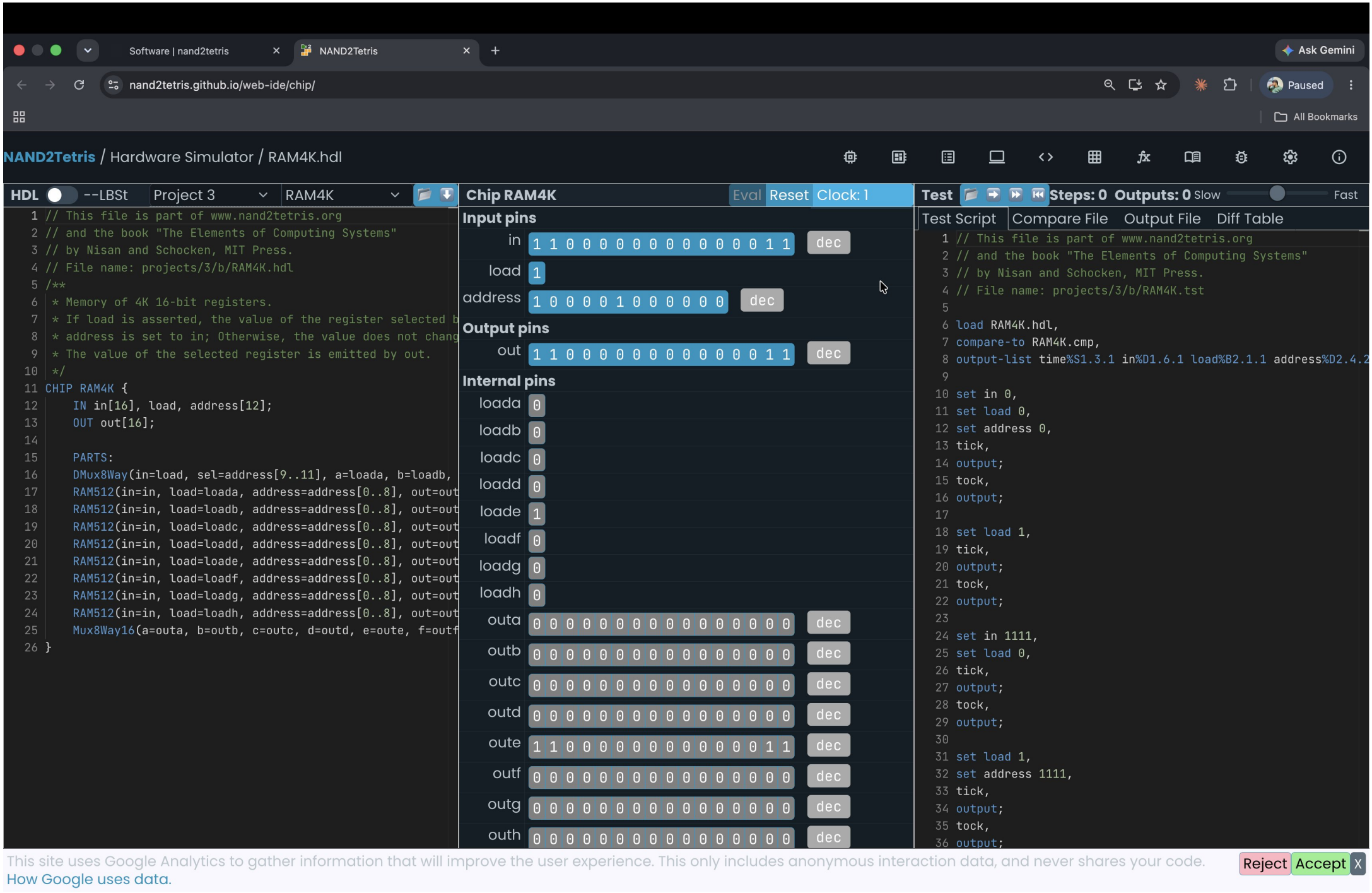

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NAND2Tetris / Hardware Simulator / Bit.hdl
HDL --LBSt Project 3 Bit
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/Bit.hdl
/**
* 1-bit register:
* If load is asserted, the register's value is set to in;
* Otherwise, the register maintains its current value:
* if (load(t)) out(t+1) = in(t), else out(t+1) = out(t)
*/
CHIP Bit {
IN in, load;
OUT out;
PARTS:
Mux(a=outLoop , b=in , sel=load , out=muxOut );
DFF(in=muxOut , out=outLoop , out=out );
}
Chip Bit Eval Reset Clock: 0+
Input pins
in 1
load 1
Output pins
out 0
Internal pins
outLoop 0
muxOut 1
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/Bit.tst
load Bit.hdl,
compare-to Bit.cmp,
output-list time%S1.4.1 in load%B1.1.2 out;
set in 0,
set load 0,
tick,
output;
tock,
output;
set in 0,
set load 1,
tick,
output;
tock,
output;
set in 1,
set load 0,
tick,
output;
tock,
output;
set in 1,
set load 1,
tick,
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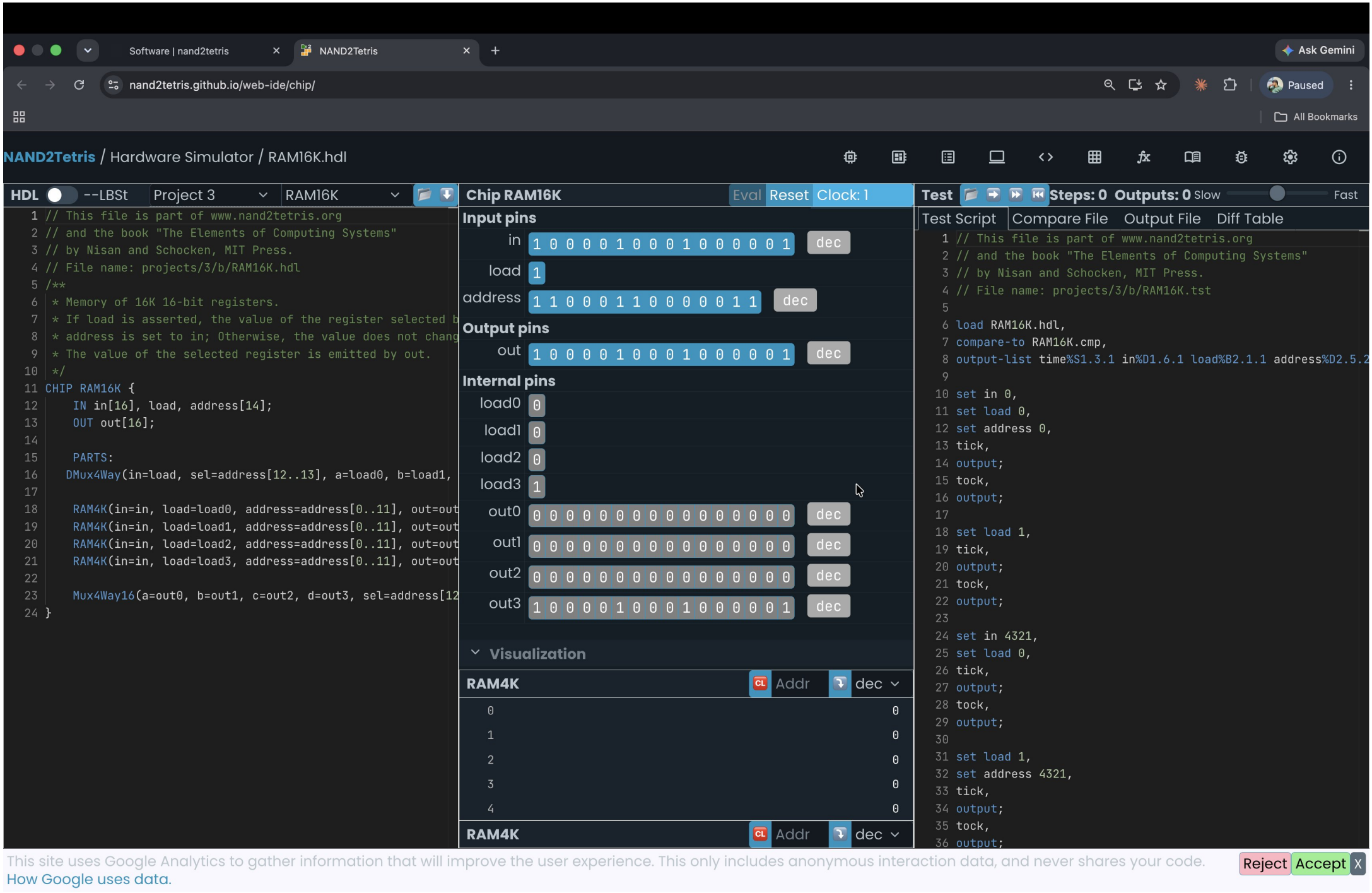

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NAND2Tetris / Hardware Simulator / Register.hdl
HDL --LBSt Project 3 Register
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/Register.hdl
/**
* 16-bit register:
* If load is asserted, the register's value is set to in;
* Otherwise, the register maintains its current value:
* if (load(t)) out(t+1) = int(t), else out(t+1) = out(t)
*/
CHIP Register {
IN in[16], load;
OUT out[16];
PARTS:
Bit(in=in[0], load=load, out=out[0]);
Bit(in=in[1], load=load, out=out[1]);
Bit(in=in[2], load=load, out=out[2]);
Bit(in=in[3], load=load, out=out[3]);
Bit(in=in[4], load=load, out=out[4]);
Bit(in=in[5], load=load, out=out[5]);
Bit(in=in[6], load=load, out=out[6]);
Bit(in=in[7], load=load, out=out[7]);
Bit(in=in[8], load=load, out=out[8]);
Bit(in=in[9], load=load, out=out[9]);
Bit(in=in[10], load=load, out=out[10]);
Bit(in=in[11], load=load, out=out[11]);
Bit(in=in[12], load=load, out=out[12]);
Bit(in=in[13], load=load, out=out[13]);
Bit(in=in[14], load=load, out=out[14]);
Bit(in=in[15], load=load, out=out[15]);
}
Chip Register
Eval
Reset
Clock: 1
Input pins
in 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1
dec
load 1
Output pins
out 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1
dec
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/Register.tst
load Register.hdl,
compare-to Register.cmp,
output-list time%S1.3.1 in%D1.6.1 load%B2.1.1 out%D1.6.1;
set in 0,
set load 0,
tick,
output;
tock,
output;
set in 0,
set load 1,
tick,
output;
tock,
output;
set in -32123,
set load 0,
tick,
output;
tock,
output;
set in 11111,
set load 0,
tick,
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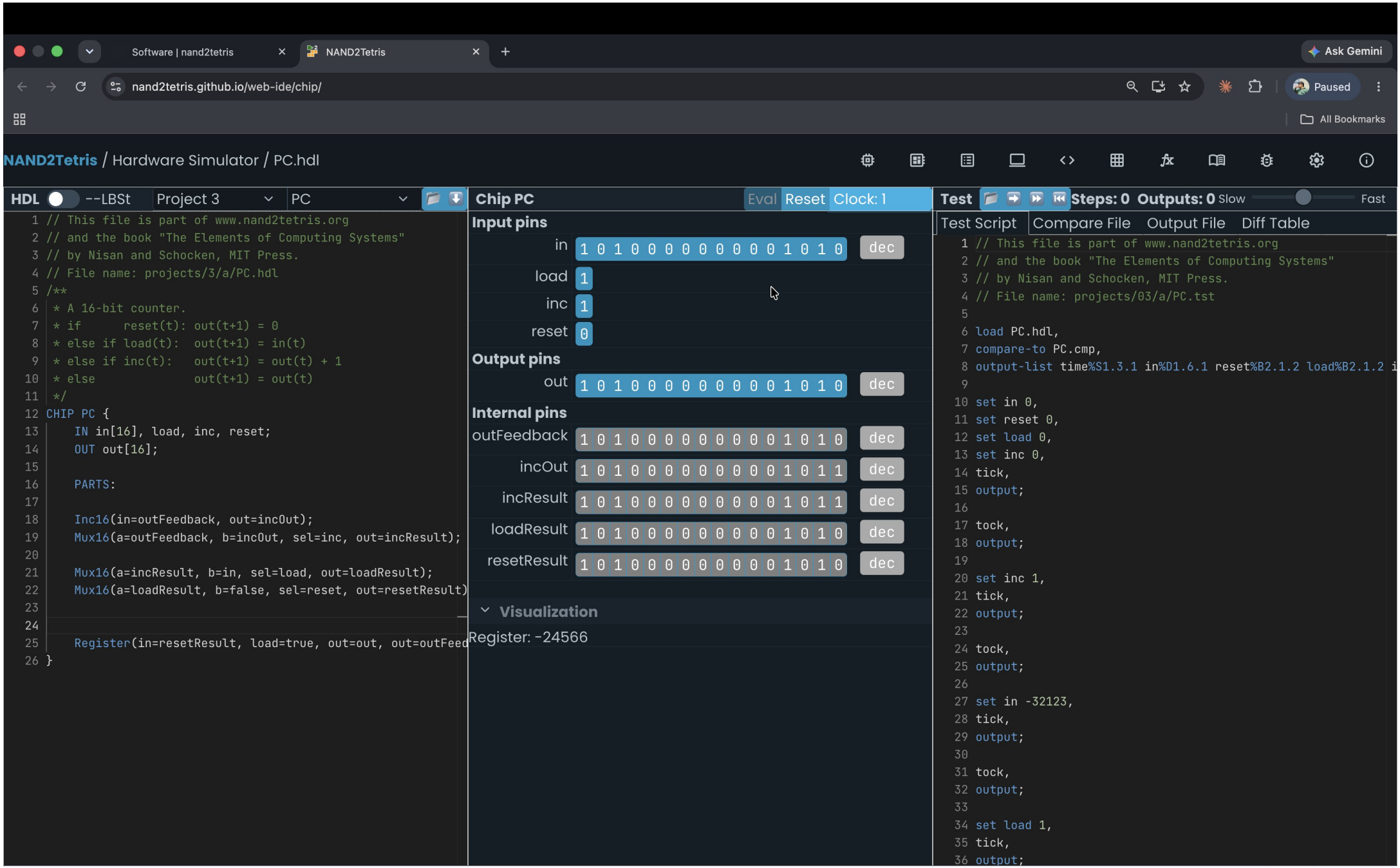

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NAND2Tetris / Hardware Simulator / RAM8.hdl
HDL --LBSt Project 3 RAM8
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/RAM8.hdl
/**
* Memory of eight 16-bit registers.
* If load is asserted, the value of the register selected b
* address is set to in; Otherwise, the value does not chang
* The value of the selected register is emitted by out.
*/
CHIP RAM8 {
IN in[16], load, address[3];
OUT out[16];
PARTS:
DMux8Way(in=load, sel=address, a=load0, b=load1, c=load2
Register(in=in, load=load0, out=out0);
Register(in=in, load=load1, out=out1);
Register(in=in, load=load2, out=out2);
Register(in=in, load=load3, out=out3);
Register(in=in, load=load4, out=out4);
Register(in=in, load=load5, out=out5);
Register(in=in, load=load6, out=out6);
Register(in=in, load=load7, out=out7);
Mux8Way16(a=out0, b=out1, c=out2, d=out3, e=out4, f=out5
}
Chip RAM8 Eval Reset Clock: 1
Input pins
in 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
load 1
address 0 1 0 dec
Output pins
out 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
Internal pins
load0 0
load1 0
load2 1
load3 0
load4 0
load5 0
load6 0
load7 0
out0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out2 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
out3 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out4 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out5 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out6 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out7 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/RAM8.tst
load RAM8.hdl,
compare-to RAM8.cmp,
output-list time%S1.3.1 in%D1.6.1 load%B2.1.1 address%D3.1.3
set in 0,
set load 0,
set address 0,
tick,
output;
tock,
output;
set load 1,
tick,
output;
tock,
output;
set in 11111,
set load 0,
tick,
output;
tock,
output;
set load 1,
set address 1,
tick,
output;
tock,
output;
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NAND2Tetris / Hardware Simulator / RAM64.hdl
HDL --LBSt Project 3 RAM64
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/RAM64.hdl
/**
* Memory of sixty four 16-bit registers.
* If load is asserted, the value of the register selected b
* address is set to in; Otherwise, the value does not chang
* The value of the selected register is emitted by out.
*/
CHIP RAM64 {
IN in[16], load, address[6];
OUT out[16];
PARTS:
DMux8Way(in=load, sel=address[3..5], a=a0, b=b0, c=c0, d
RAM8(in=in, load=a0, address=address[0..2], out=out0);
RAM8(in=in, load=b0, address=address[0..2], out=out1);
RAM8(in=in, load=c0, address=address[0..2], out=out2);
RAM8(in=in, load=d0, address=address[0..2], out=out3);
RAM8(in=in, load=e0, address=address[0..2], out=out4);
RAM8(in=in, load=f0, address=address[0..2], out=out5);
RAM8(in=in, load=g0, address=address[0..2], out=out6);
RAM8(in=in, load=h0, address=address[0..2], out=out7);
Mux8Way16(a=out0, b=out1, c=out2, d=out3, e=out4, f=out5
}
Chip RAM64 Eval Reset Clock: 1
Input pins
in 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
load 1
address 0 0 1 1 0 0 dec
Output pins
out 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
Internal pins
a0 0
b0 1
c0 0
d0 0
e0 0
f0 0
g0 0
h0 0
out0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
out2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out3 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out4 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out5 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out6 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out7 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/RAM64.tst
load RAM64.hdl,
compare-to RAM64.cmp,
output-list time%S1.3.1 in%D1.6.1 load%B2.1.1 address%D2.3.2
set in 0,
set load 0,
set address 0,
tick,
output;
tock,
output;
set load 1,
tick,
output;
tock,
output;
set in 1313,
set load 0,
tick,
output;
tock,
output;
set load 1,
set address 13,
tick,
output;
tock,
output;
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NAND2Tetris / Hardware Simulator / RAM512.hdl
HDL --LBSt Project 3 RAM512
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/b/RAM512.hdl
/**
* Memory of 512 16-bit registers.
* If load is asserted, the value of the register selected b
* address is set to in; Otherwise, the value does not chang
* The value of the selected register is emitted by out.
*/
CHIP RAM512 {
IN in[16], load, address[9];
OUT out[16];
PARTS:
DMux8Way(in=load, sel=address[6..8], a=la, b=lb, c=lc, d
RAM64(in=in, load=la, address=address[0..5], out=oa);
RAM64(in=in, load=lb, address=address[0..5], out=ob);
RAM64(in=in, load=lc, address=address[0..5], out=oc);
RAM64(in=in, load=ld, address=address[0..5], out=od);
RAM64(in=in, load=le, address=address[0..5], out=oe);
RAM64(in=in, load=lf, address=address[0..5], out=of);
RAM64(in=in, load=lg, address=address[0..5], out=og);
RAM64(in=in, load=lh, address=address[0..5], out=oh);
Mux8Way16(a=oa, b=ob, c=oc, d=od, e=oe, f=of, g=og, h=oh
}
Chip RAM512 Eval Reset Clock: 1
Input pins
in 1 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 dec
load 1
address 1 0 0 0 0 0 0 0 1 dec
Output pins
out 1 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 dec
Internal pins
la 0
lb 0
lc 0
ld 0
le 1
lf 0
lg 0
lh 0
oa 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
ob 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
oc 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
od 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
oe 1 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 dec
of 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
og 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
oh 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/b/RAM512.tst
load RAM512.hdl,
compare-to RAM512.cmp,
output-list time%S1.3.1 in%D1.6.1 load%B2.1.1 address%D2.3.2
set in 0,
set load 0,
set address 0,
tick,
output;
tock,
output;
set load 1,
tick,
output;
tock,
output;
set in 13099,
set load 0,
tick,
output;
tock,
output;
set load 1,
set address 130,
tick,
output;
tock,
output;
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NAND2Tetris / Hardware Simulator / RAM4K.hdl
HDL --LBSt Project 3 RAM4K
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/b/RAM4K.hdl
/**
* Memory of 4K 16-bit registers.
* If load is asserted, the value of the register selected b
* address is set to in; Otherwise, the value does not chang
* The value of the selected register is emitted by out.
*/
CHIP RAM4K {
IN in[16], load, address[12];
OUT out[16];
PARTS:
DMux8Way(in=load, sel=address[9..11], a=loada, b=loadb,
RAM512(in=in, load=loada, address=address[0..8], out=out
RAM512(in=in, load=loadb, address=address[0..8], out=out
RAM512(in=in, load=loadc, address=address[0..8], out=out
RAM512(in=in, load=loadd, address=address[0..8], out=out
RAM512(in=in, load=loade, address=address[0..8], out=out
RAM512(in=in, load=loadf, address=address[0..8], out=out
RAM512(in=in, load=loadg, address=address[0..8], out=out
RAM512(in=in, load=loadh, address=address[0..8], out=out
Mux8Way16(a=outa, b=outb, c=outc, d=outd, e=oute, f=outf
}
Chip RAM4K Eval Reset Clock: 1
Input pins
in 1 1 0 0 0 0 0 0 0 0 0 0 0 0 1 1 dec
load 1
address 1 0 0 0 0 1 0 0 0 0 0 0 dec
Output pins
out 1 1 0 0 0 0 0 0 0 0 0 0 0 0 1 1 dec
Internal pins
loada 0
loadb 0
loadc 0
loadd 0
loade 1
loadf 0
loadg 0
loadh 0
outa 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
outb 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
outc 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
outd 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
oute 1 1 0 0 0 0 0 0 0 0 0 0 0 0 1 1 dec
outf 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
outg 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
outh 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/b/RAM4K.tst
load RAM4K.hdl,
compare-to RAM4K.cmp,
output-list time%S1.3.1 in%D1.6.1 load%B2.1.1 address%D2.4.2
set in 0,
set load 0,
set address 0,
tick,
output;
tock,
output;
set load 1,
tick,
output;
tock,
output;
set in 1111,
set load 0,
tick,
output;
tock,
output;
set load 1,
set address 1111,
tick,
output;
tock,
output;
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NAND2Tetris / Hardware Simulator / RAM16K.hdl
HDL --LBSt Project 3 RAM16K
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// by Nisan and Schocken, MIT Press.
// File name: projects/3/b/RAM16K.hdl
/**
* Memory of 16K 16-bit registers.
* If load is asserted, the value of the register selected b
* address is set to in; Otherwise, the value does not chang
* The value of the selected register is emitted by out.
*/
CHIP RAM16K {
IN in[16], load, address[14];
OUT out[16];
PARTS:
DMux4Way(in=load, sel=address[12..13], a=load0, b=load1,
RAM4K(in=in, load=load0, address=address[0..11], out=out
RAM4K(in=in, load=load1, address=address[0..11], out=out
RAM4K(in=in, load=load2, address=address[0..11], out=out
RAM4K(in=in, load=load3, address=address[0..11], out=out
Mux4Way16(a=out0, b=out1, c=out2, d=out3, sel=address[12
}
Chip RAM16K Eval Reset Clock: 1
Input pins
in 1 0 0 0 0 1 0 0 0 1 0 0 0 0 0 1 dec
load 1
address 1 1 0 0 0 1 1 0 0 0 0 0 1 1 dec
Output pins
out 1 0 0 0 0 1 0 0 0 1 0 0 0 0 0 1 dec
Internal pins
load0 0
load1 0
load2 0
load3 1
out0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out3 1 0 0 0 0 1 0 0 0 1 0 0 0 0 0 1 dec
Visualization
RAM4K CL Addr dec
0 0
1 0
2 0
3 0
4 0
RAM4K CL Addr dec
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/b/RAM16K.tst
load RAM16K.hdl,
compare-to RAM16K.cmp,
output-list time%S1.3.1 in%D1.6.1 load%B2.1.1 address%D2.5.2
set in 0,
set load 0,
set address 0,
tick,
output;
tock,
output;
set load 1,
tick,
output;
tock,
output;
set in 4321,
set load 0,
tick,
output;
tock,
output;
set load 1,
set address 4321,
tick,
output;
tock,
output;
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All Bookmarks
NAND2Tetris / Hardware Simulator / PC.hdl
HDL --LBSt Project 3 PC
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/PC.hdl
/**
* A 16-bit counter.
* if reset(t): out(t+1) = 0
* else if load(t): out(t+1) = in(t)
* else if inc(t): out(t+1) = out(t) + 1
* else out(t+1) = out(t)
*/
CHIP PC {
IN in[16], load, inc, reset;
OUT out[16];
PARTS:
Inc16(in=outFeedback, out=incOut);
Mux16(a=outFeedback, b=incOut, sel=inc, out=incResult);
Mux16(a=incResult, b=in, sel=load, out=loadResult);
Mux16(a=loadResult, b=false, sel=reset, out=resetResult)
Register(in=resetResult, load=true, out=out, out=outFeed
}
Chip PC
Eval
Reset
Clock: 1
Input pins
in 1 0 1 0 0 0 0 0 0 0 0 0 1 0 1 0 dec
load 1
inc 1
reset 0
Output pins
out 1 0 1 0 0 0 0 0 0 0 0 0 1 0 1 0 dec
Internal pins
outFeedback 1 0 1 0 0 0 0 0 0 0 0 0 1 0 1 0 dec
incOut 1 0 1 0 0 0 0 0 0 0 0 0 1 0 1 1 dec
incResult 1 0 1 0 0 0 0 0 0 0 0 0 1 0 1 1 dec
loadResult 1 0 1 0 0 0 0 0 0 0 0 0 1 0 1 0 dec
resetResult 1 0 1 0 0 0 0 0 0 0 0 0 1 0 1 0 dec
Visualization
Register: -24566
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/03/a/PC.tst
load PC.hdl,
compare-to PC.cmp,
output-list time%S1.3.1 in%D1.6.1 reset%B2.1.2 load%B2.1.2 i
set in 0,
set reset 0,
set load 0,
set inc 0,
tick,
output;
tock,
output;
set inc 1,
tick,
output;
tock,
output;
set in -32123,
tick,
output;
tock,
output;
set load 1,
tick,
output;
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# Logic Design Overview

NAND2Tetris builds complex hardware compositionally. The basic logic relationships below explain the main structures used in the HDL listings.

| Block | Logic relationship | Inputs → Output |
|---|---|---|
| NOT | out = ¬a | a → out |
| AND | out = a · b | a,b → out |
| OR | out = a + b | a,b → out |
| XOR | out = a ⊕ b | a,b → out |
| MUX | out = (¬sel·a) + (sel·b) | a,b,sel → out |
| ALU | controlled AND / ADD pipeline | x,y + 6 control bits → out,zr,ng |
| PC | state register with reset/load/inc | in + controls → out |

NAND universality chain: NAND → NOT → AND / OR → MUX / DMUX → 16-bit gates → ALU and sequential components.

# ALU Function Table

| zx | nx | zy | ny | f | no | Function |
|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 0 | 1 | 1 | -1 |
| 1 | 0 | 1 | 1 | 0 | 0 | x |
| 0 | 1 | 1 | 1 | 0 | 0 | x |
| 1 | 1 | 1 | 0 | 0 | 0 | !x |
| 1 | 0 | 1 | 1 | 0 | 0 | x+1 |
| 1 | 1 | 1 | 1 | 0 | 0 | x-1 |
| 1 | 0 | 1 | 1 | 1 | 0 | y |
| 1 | 1 | 1 | 1 | 1 | 1 | !y |
| 1 | 0 | 0 | 1 | 1 | 0 | y+1 |
| 1 | 1 | 0 | 1 | 1 | 1 | y-1 |
| 0 | 0 | 1 | 1 | 1 | 1 | x+y |
| 0 | 1 | 0 | 1 | 1 | 1 | x-y |
| 0 | 0 | 1 | 1 | 0 | 0 | x&y |
| 0 | 1 | 0 | 1 | 0 | 1 | x\|y |

The six control bits determine preprocessing of x and y, whether the operation is AND or ADD, and whether the final result is negated.

## Reflection — NAND Universality

This work made NAND universality practical rather than just theoretical. A NAND gate can be used to construct NOT, AND and OR, and those gates can then be combined into multiplexers, demultiplexers, 16-bit components and the ALU. The HDL exercises also showed that hardware design is strongly dependent on exact wiring, bus widths, selector order and control priority. The Hardware Simulator provides a useful bridge between the logical design and observable behavior because every chip can be tested against a truth table or test script. Overall, the project demonstrated how a small primitive gate can serve as the foundation for much larger digital systems.